



THE LAB CODER

Intermediate Level (Grove Hat)

Updated 15 Feb 2026

Content

Page	Unit	Topic
3	3017	<u>Variables & Math Operators</u>
13	3018	<u>Variables & Comparison Operators</u>
20	3019	<u>String Variables</u>
27	3020	<u>Count Loop</u>
34	3021	<u>Intermediate 1 Project</u>
36	3022	<u>List Basics</u>
45	3023	<u>List Updating</u>
55	3024	<u>List Manipulation</u>
64	3025	<u>Function</u>
72	3026	<u>Function Parameter</u>
82	3027	<u>Function Return</u>
88	3028	<u>Function Applications</u>
90	3029	<u>Intermediate 2 Project</u>

3017 - Variables & Math Operators

Learning

A variable is used to store a value that can be reused and updated throughout a program. This simple yet powerful concept enables programmers to move beyond writing small, rigid scripts to building large, dynamic, and scalable programs.

In this unit, you'll learn how to work with variables and create formulas using them.

In class, we will explore the following:

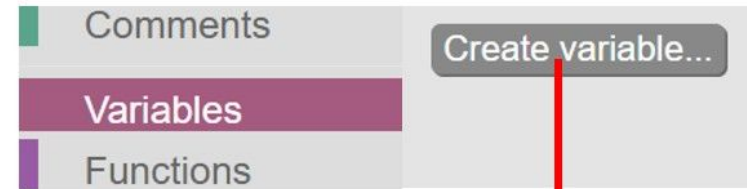
- Create and modify variable values.
- Create formulas using Math blocks.
- Application including a Math Times Tables quiz program.



2017 - Variables & Math Operators

Creating variable

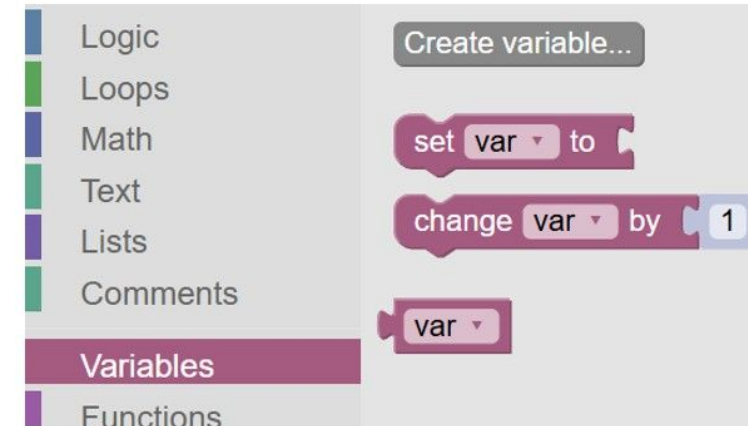
A variable is used to store a value that will be used subsequently. When you can create a variable, give it a meaningful name (what it stores, e.g. name, age, superpower, etc).




A dialog box titled 'web.thelab.sg says' with the text 'New variable name:'. Below this is a text input field containing the text 'var'. A red arrow points to the input field with the text 'key in your variable name here'.

key in your variable name here

Once a variable is created, it will show up in the menu.



3017 - Variables & Math Operators

Setting and Modifying variable

A variable can store either a number (eg. 1, 9, 22, 5.47, etc), a text (eg. "ABC") or other **data type**. It stores only one value at any time.

It is important to know what data type the variable is storing so that you can perform the right operation. Below shows some examples using the **change block** and **create-text block**.

Assign a **text** to the variable



Assign a **number** to the variable



Add a **number** to the variable



This only works when the variable is currently storing a **number** (not text or empty)

Append values behind the current value



this block joins texts/numbers into one row of text

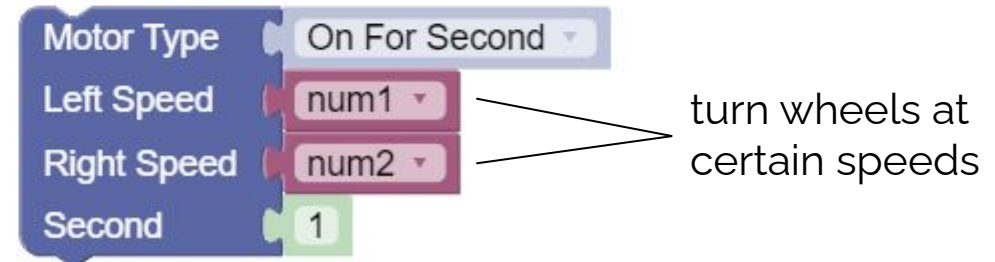
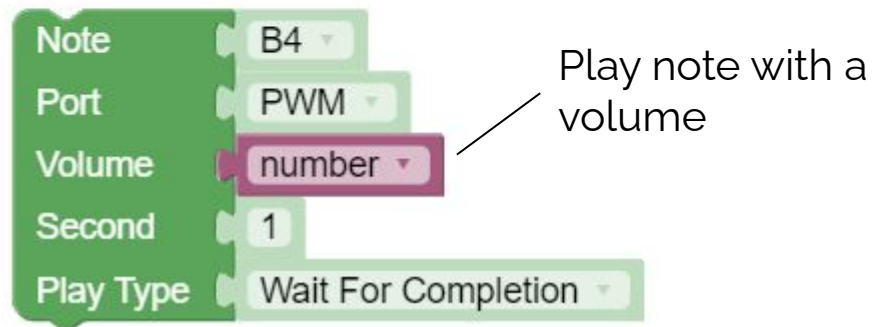
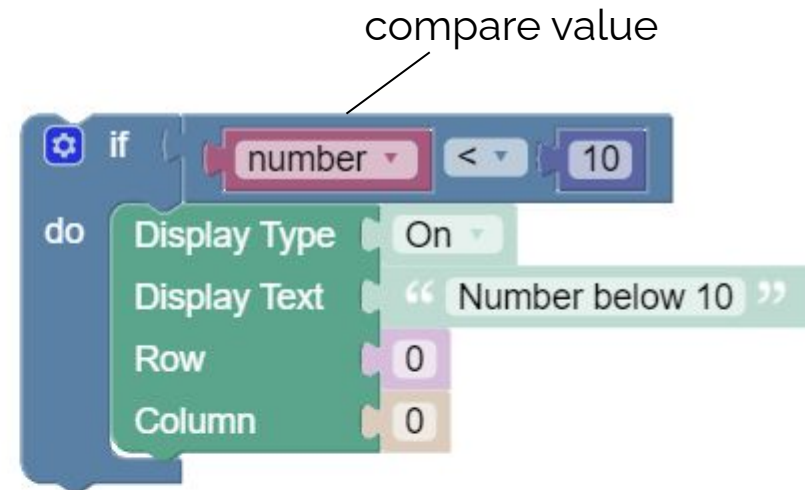
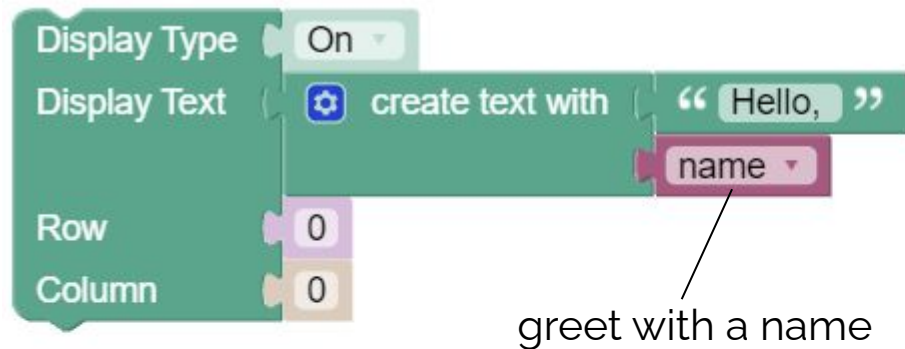
Add a **negative number** decreases the variable value



3017 - Variables & Math Operators

Using Variable

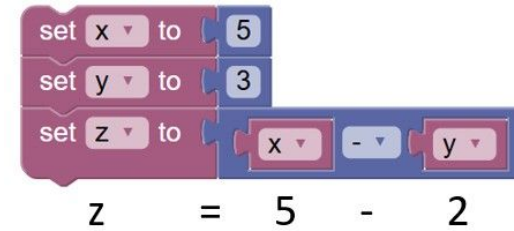
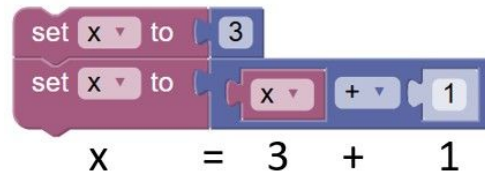
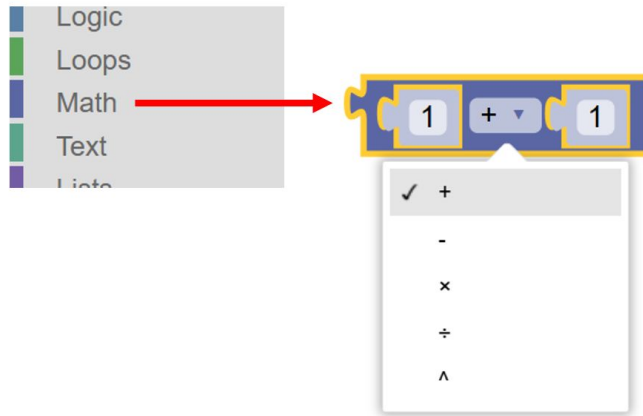
Variables can be used in many ways. Below shows some examples.



3017 - Variables & Math Operators

Math Operators

The Math operators are used to perform mathematical operations with the variables. Below shows some examples.



When using multiple Math operator blocks to create formula, be careful how you place them together.



Exercise Q1

The LCD screen has 2 rows (numbered 0 to 1) and 16 columns (numbered 0 to 15). Which of the following will display an "X" moving **from the right most column to the left most column** of the LCD screen?

Diagram 1

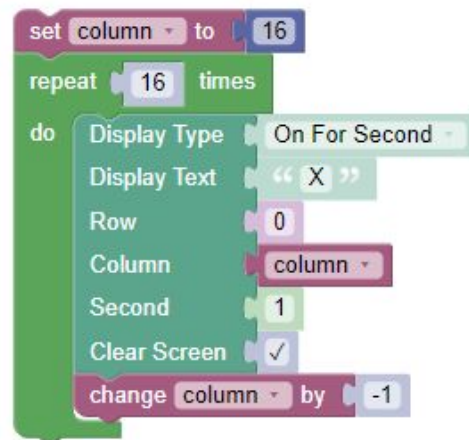


Diagram 3

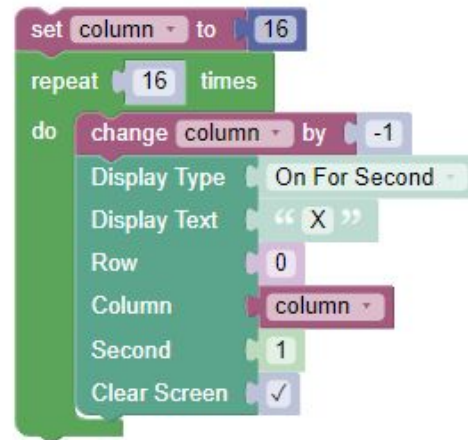


Diagram 2

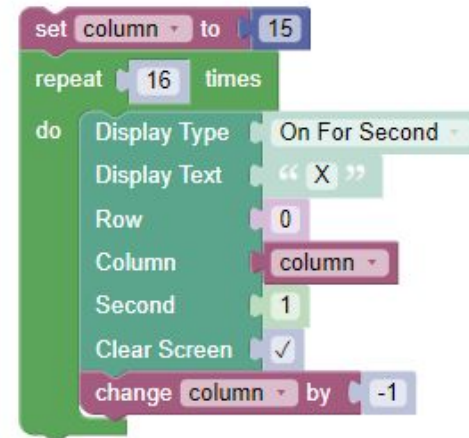
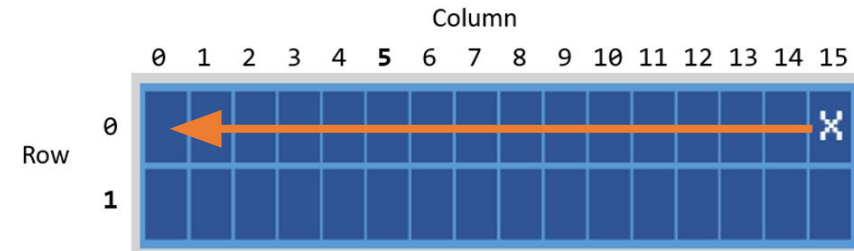
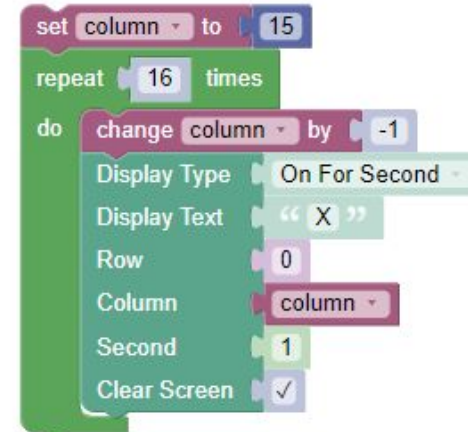


Diagram 4



- A) Diagram 1 only
- B) Diagram 2 only
- C) Diagrams 1 and 4 only
- D) Diagrams 2 and 3 only

Exercise Q2

The formula for BMI is:

$$\text{bmi} = \text{weight} \div (\text{height} \times \text{height})$$

Which of the code snippets below can be a correct representation?

Diagram 1

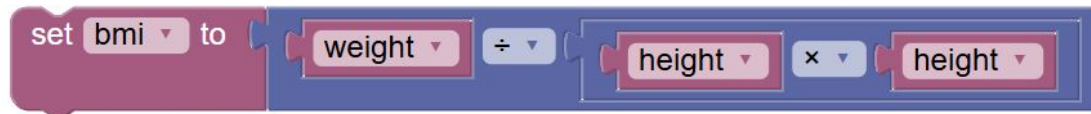
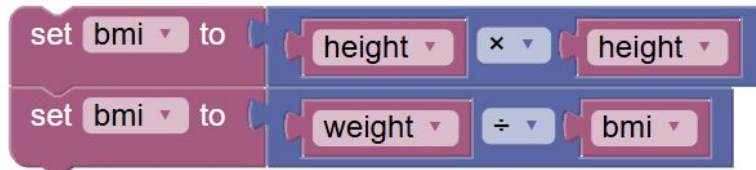


Diagram 2



Diagram 3



- A) Diagram 1 only
- B) Diagram 2 only
- C) Diagrams 1 and 2 only
- D) Diagrams 1 and 3 only

Exercise Q3

Which of the following code snippets will **always display 5** when the condition of the If statement is true?

- A) Diagram 1 only
- B) Diagram 2 only
- C) Diagram 3 only
- D) Diagram 1 and 3 only

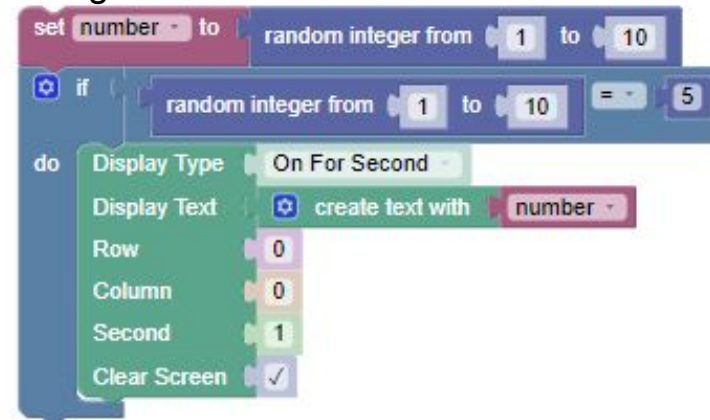
Diagram 1



Diagram 2



Diagram 3



Exercise Q4

Create a code to display a variable value on the LCD.

The variable begins with value 0.

Whenever, D22 is pressed and released, increase the variable by 1.

Whenever, D24 is pressed and released, decrease the variable by 1.

Answers

Q1: D

The column must run from 15 to 0 when the LCD block is executed. Diagram 1 counts from 16 while diagram 4 counts from 14.

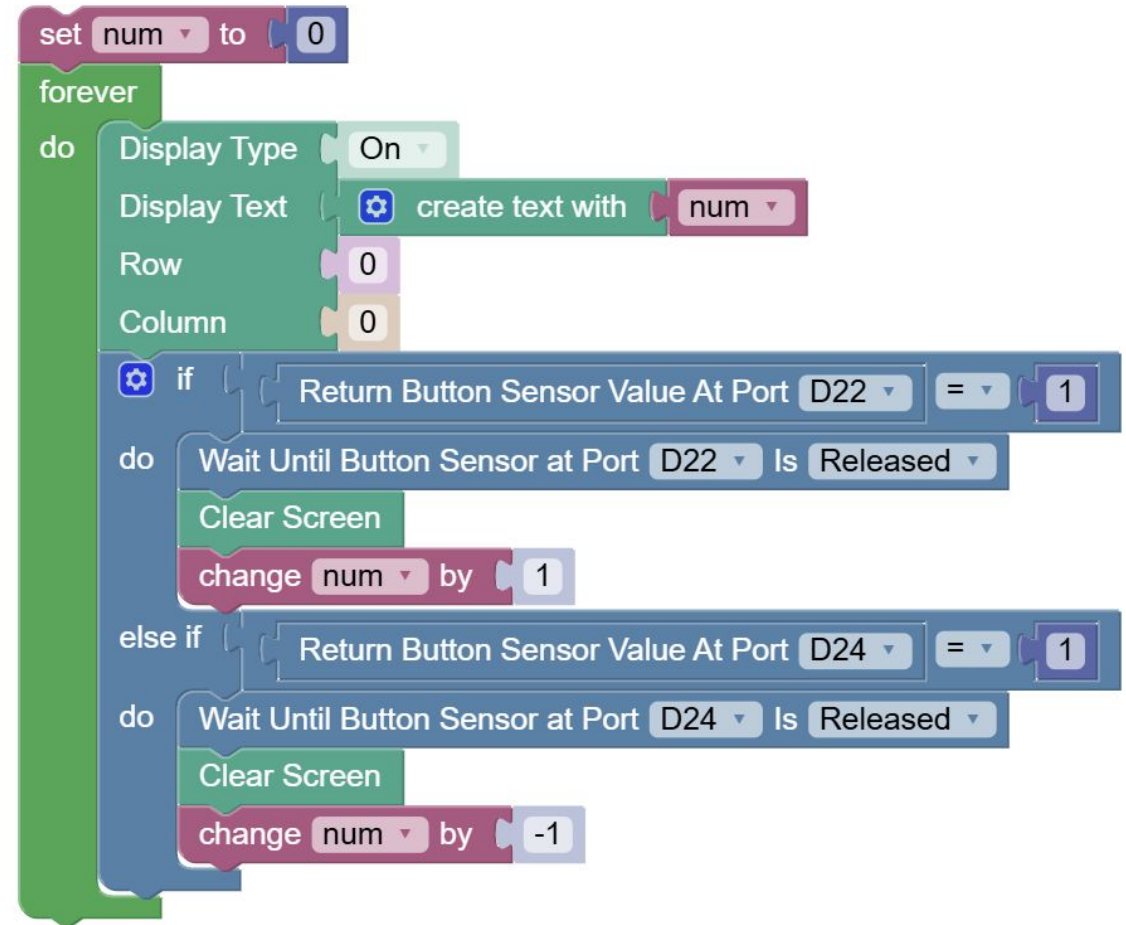
Q2: D

Diagram 2 represents
 $\text{bmi} = (\text{weight}/\text{height}) \times \text{height}$

Q3: B

Each time the random block is used, a new random number will be generated. For diagrams 1 and 3, the 2 random blocks may give different values.

Q4:



3018 - Variables & Comparison Operators

Learning

Variables drive decisions in code, often through **comparisons** like greater than, less than, equal to, etc. One of the most common patterns is **range checking**, that is, ensuring values stay within valid boundaries to prevent errors and keep logic on track.

In class, we will explore the following:

- Practise using comparison operators with variables.
- Learn about the interesting concept of Time Blindness and develop a simple “diagnostic tool”.
- Applying range detection to develop fun applications including a Sumo Wrestling game.
- More practices on breaking down problem with flow diagrams.



3018 - Variables & Comparison Operators

Comparison Operators

Here are some examples of **variable comparison conditions**.

Condition: variable equals to "text"

Returns True/False ←



Condition: variable more than 10

Returns True/False ←



Condition: variable less than or equals to 100

Returns True/False ←



Condition: variable num1 not equal
to variable num2

Returns True/False ←



Exercise Q1

Which of the following code snippets will display the sequence 1, 2, 3, 1, 2, 3, 1, 2, 3 ... when you keep pressing the button?

Diagram 1

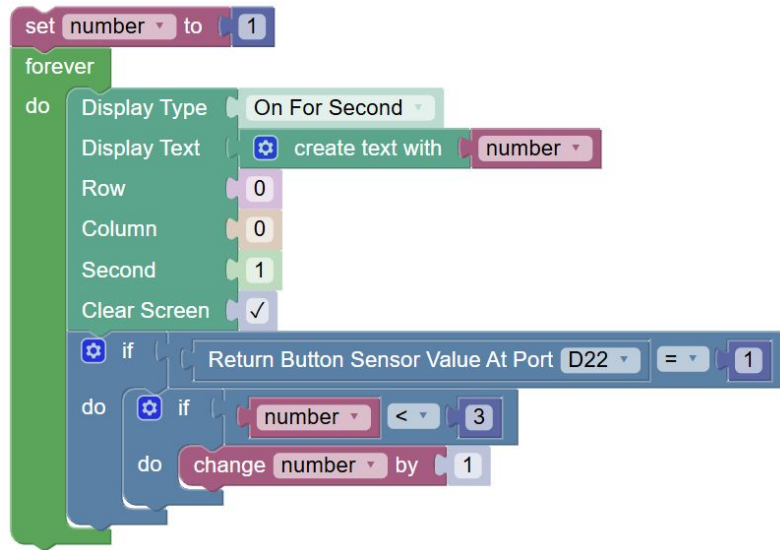


Diagram 2

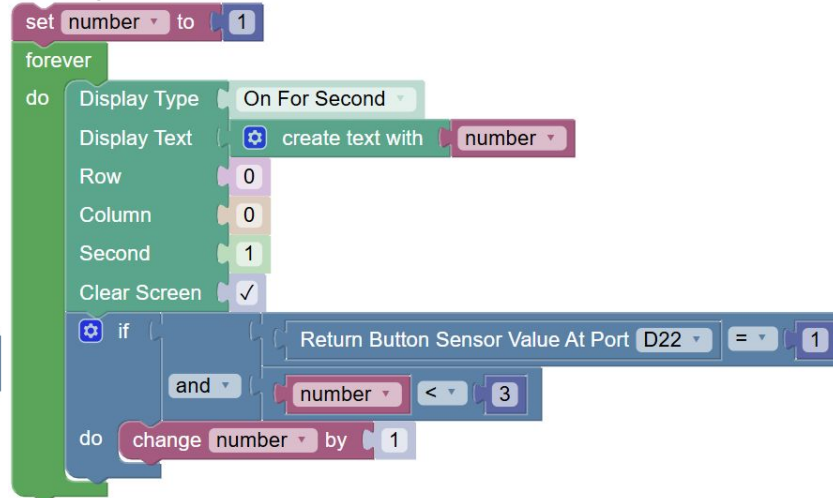
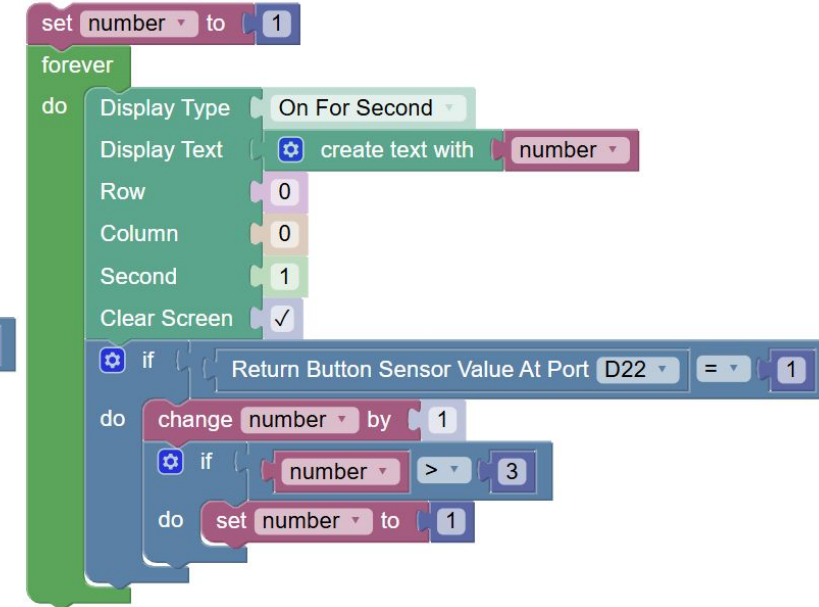


Diagram 3



- A) Diagram 1 only
- B) Diagram 2 only
- C) Diagram 3 only
- D) All of the above

Exercise Q2

Which of the following can be the condition to check that num1, num2 and num3 are in ascending order?

Diagram 1



Diagram 2



Diagram 3



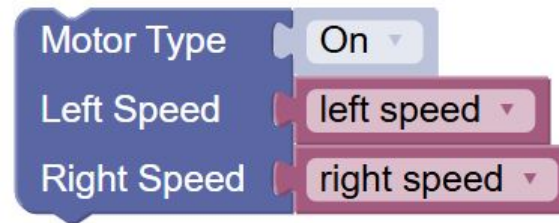
Diagram 4



- A) Diagram 1
- B) Diagram 2
- C) Diagram 3
- D) Diagram 4

Exercise Q3

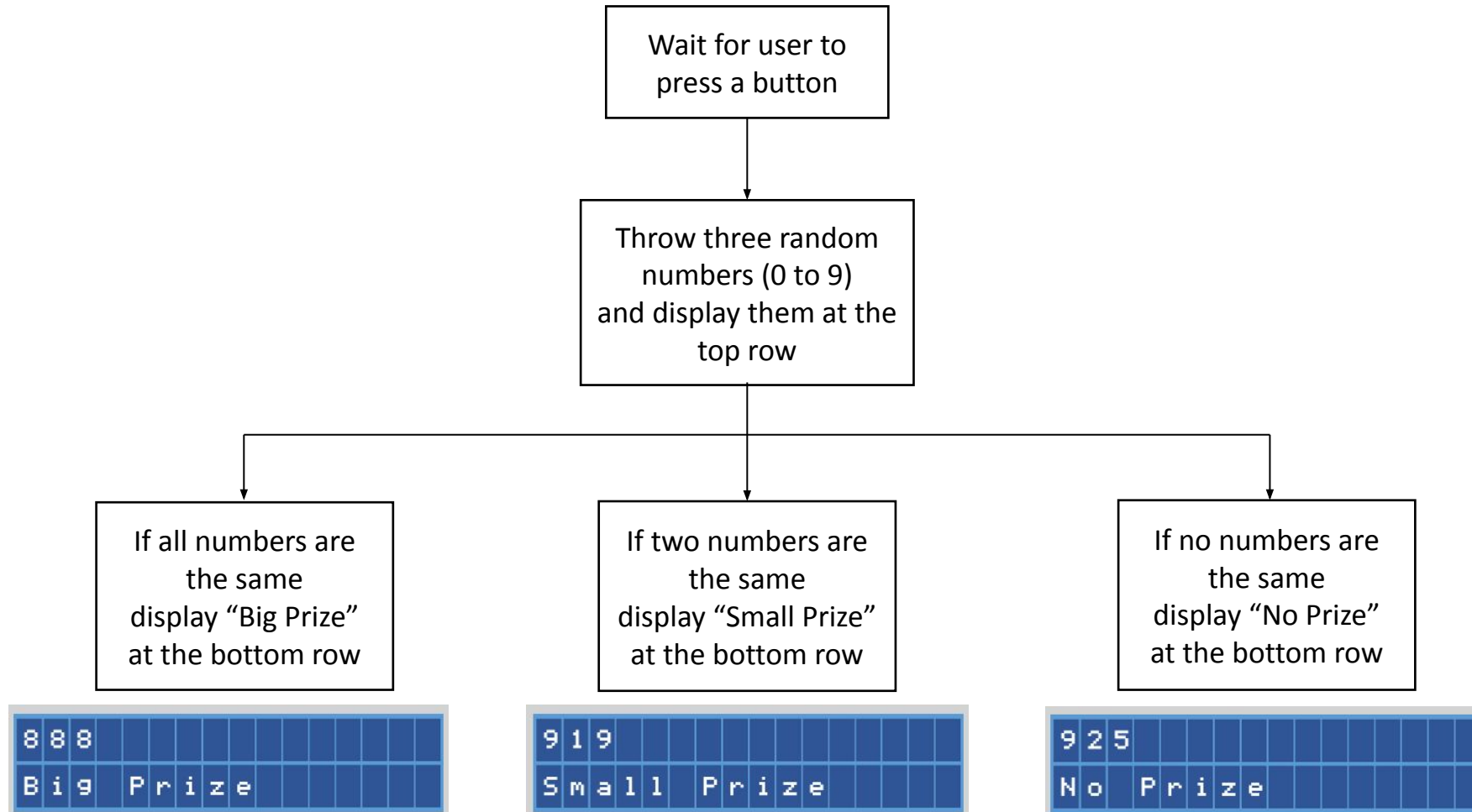
The motor of a robot car is controlled by 2 variables as shown:



Write a program to set the variables to random values from 10 to 100 whenever a button is pressed and released. The robot car will move continuously using the variables.

Exercise Q4

Create a lucky spin program with the following logic control.



Answers

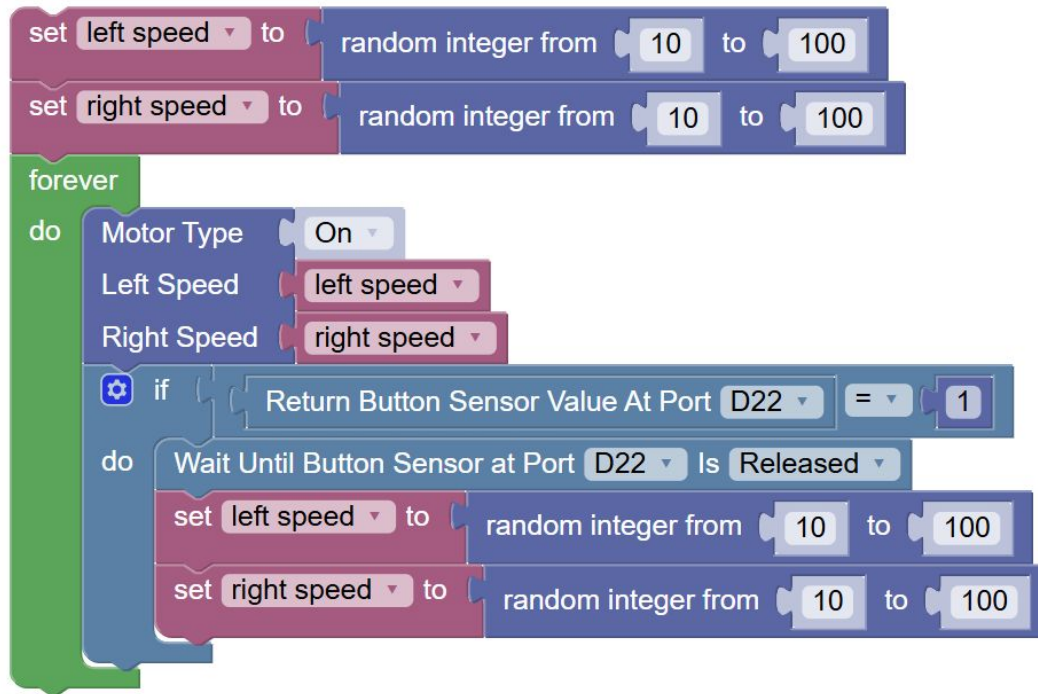
Q1: C

Diagrams 1 and 2 will display
1, 2, 3, 3, 3, 3, ...

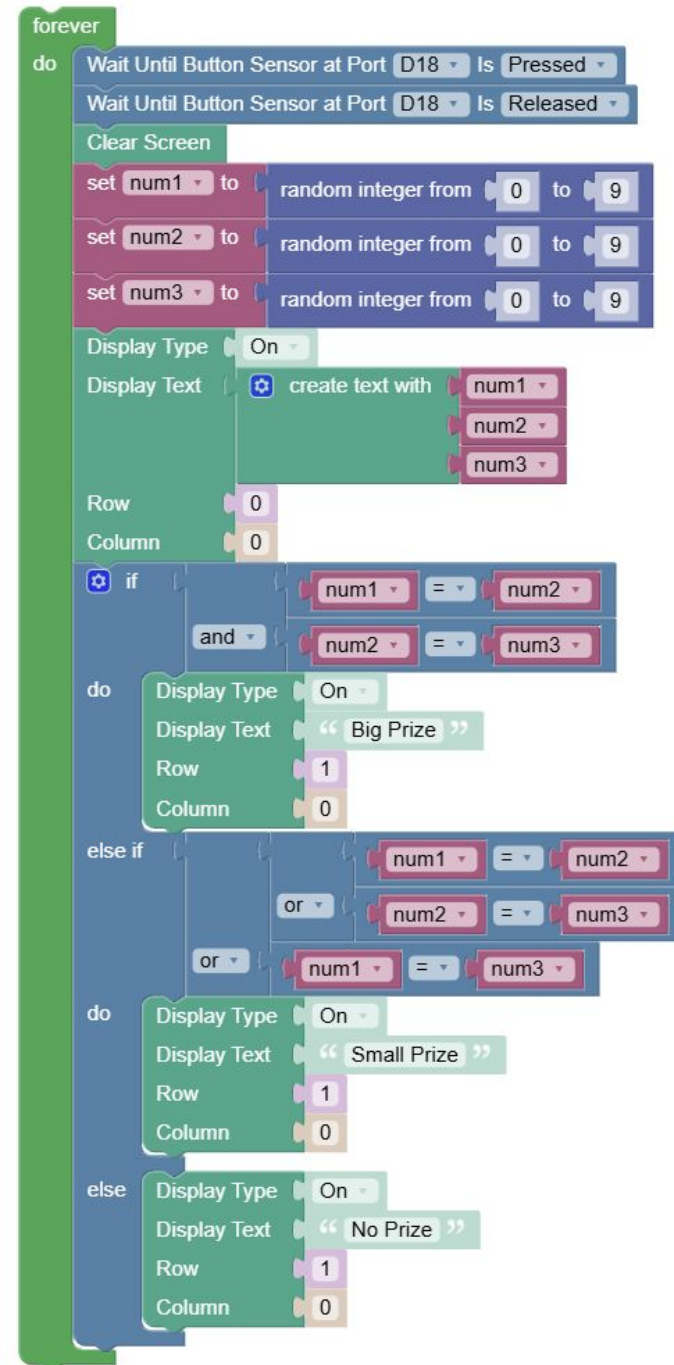
Q2: B

For ascending order, each
number is smaller than the
next.

Q3:



Q4:



3019 - String Variables

Learning

Variables don't just hold numbers, they can also store **text**, which we call **strings** in computing. Learning how to manipulate strings is a core coding skill and often a favorite challenge in programming competitions.

In class, we will explore the following:

- Learn more about modifying string variables.
- Strengthen your skills in breaking down problem using flow diagram, data table, and output analysis, applied through the development of a classic Scissors-Paper-Stone game (Player vs. Robot edition).
- Discover how programming intersects with linguistics through the development of a fascinating application that generates imaginative, fictional names.



3019 - String Variables

String variables

Here are some examples of **joining values into a string**.

If distance is 10, then var will contain "Distance: 10cm"



Exercise Q1

Which of the following code snippets will display "Hello, world!" in a **single row**?

Diagram 1

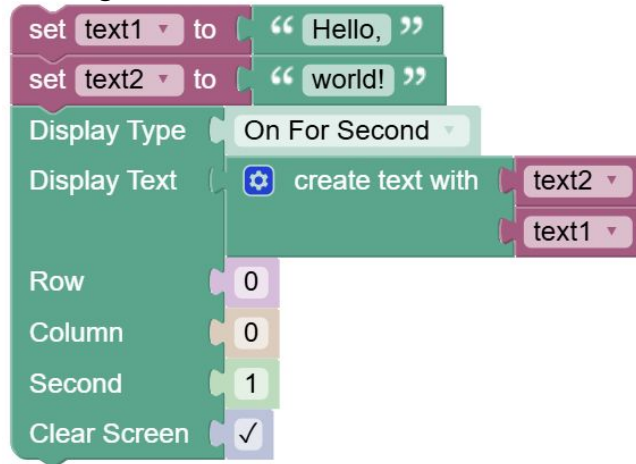


Diagram 3

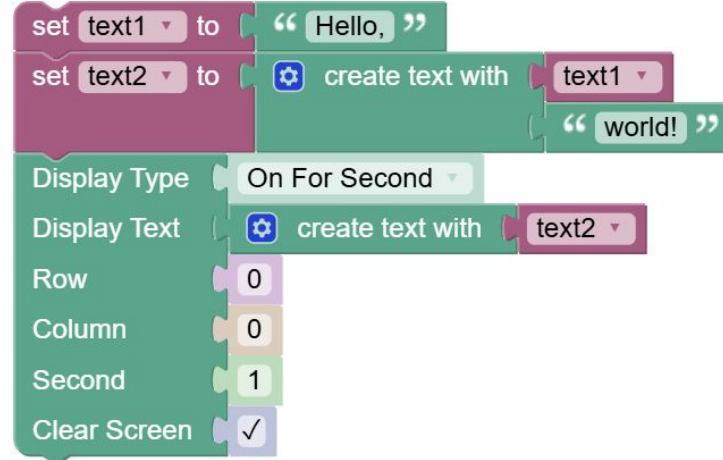


Diagram 2

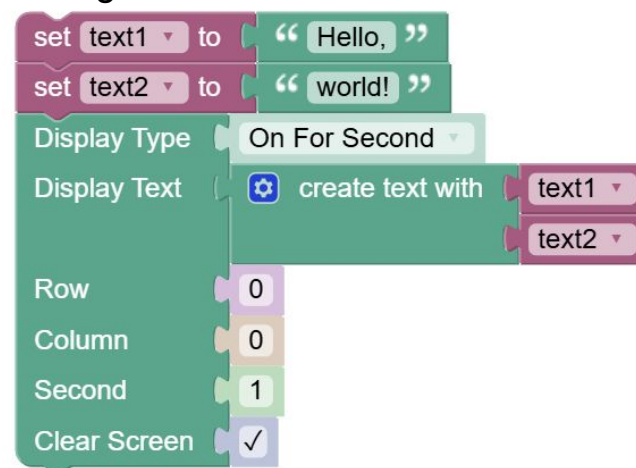
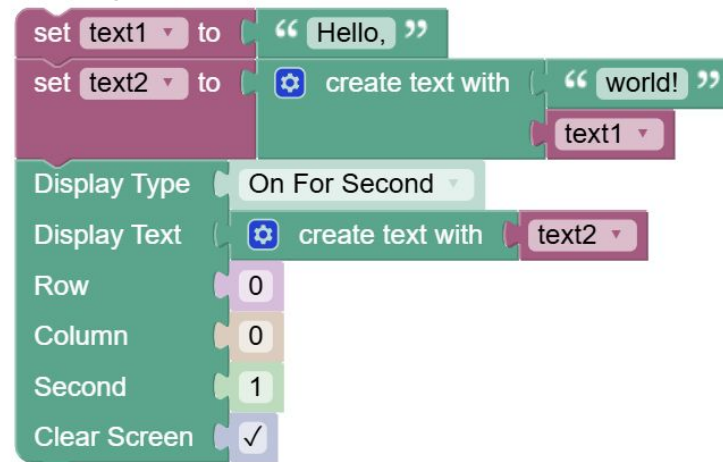


Diagram 4



- A) Diagram 1 only
- B) Diagram 2 only
- C) Diagram 1 and 4 only
- D) Diagrams 2 and 3 only

Exercise Q2

You want to display a "#" that grows by one character every second until it becomes "#####" (16 "#s). Which of the following code snippets will achieve that?

		Column															
Row		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	0																
	1																

- A) Diagram 1 only
- B) Diagram 2 only
- C) Diagram 3 only
- D) Diagrams 1, 2 and 3

Diagram 1

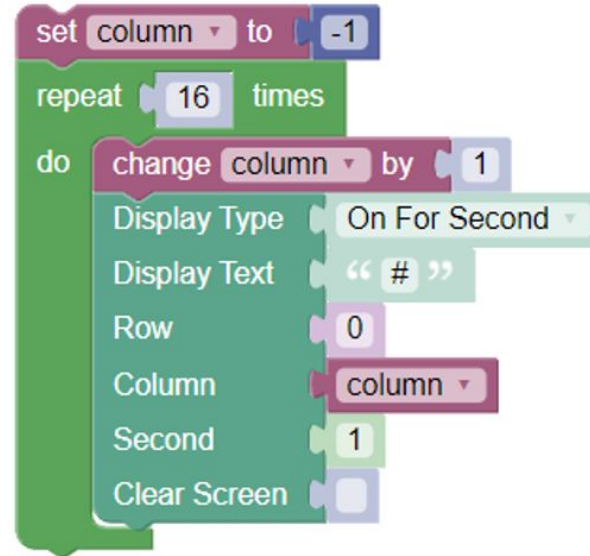
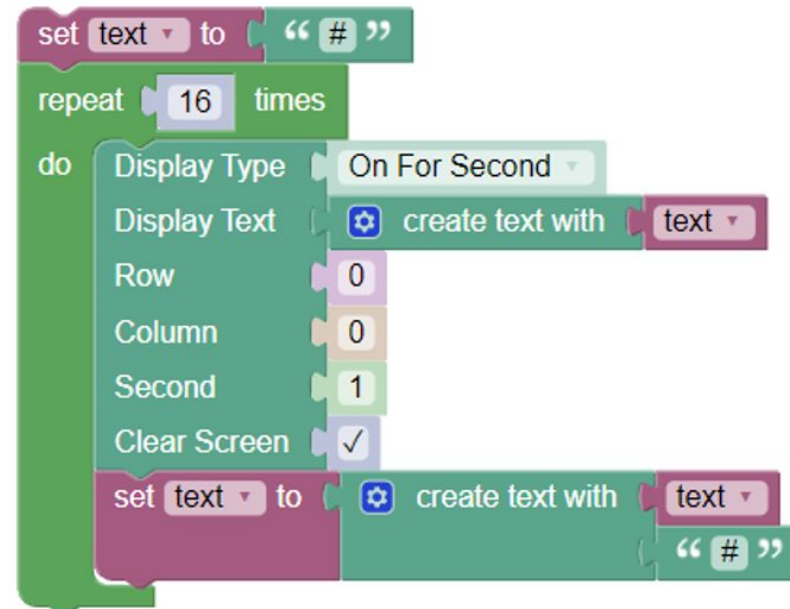
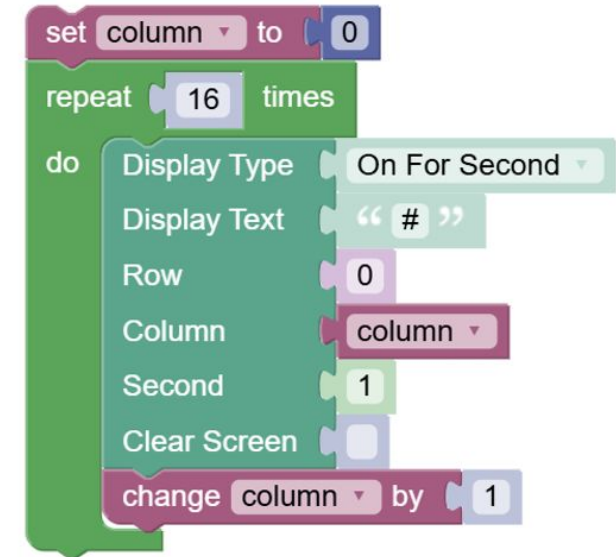
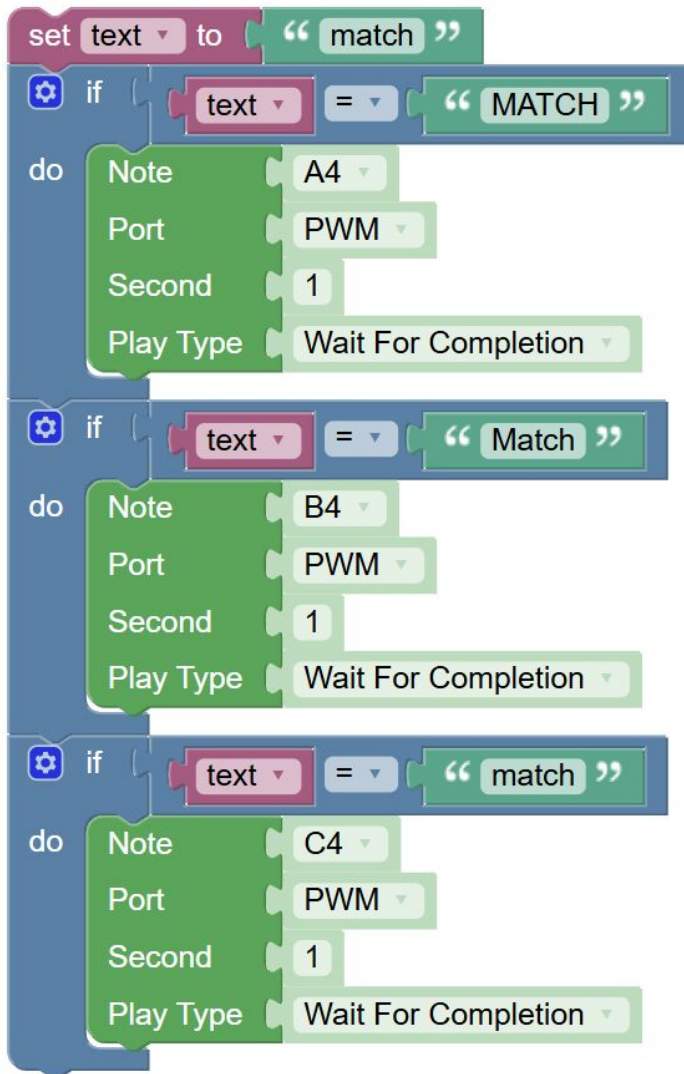


Diagram 3



Exercise Q3

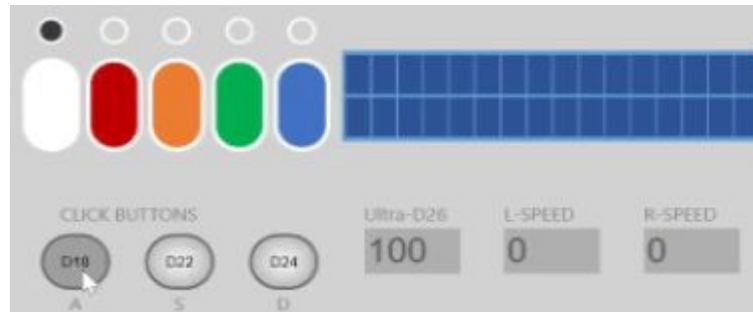
Which note will be played when the code is run?



- A) A4
- B) B4
- C) C4
- D) All the notes

Exercise Q4

Using a string variable, write a code that adds a “|” character to it whenever Button D22 is pressed and released. The clip below shows it execution.



Answers

Q1: D

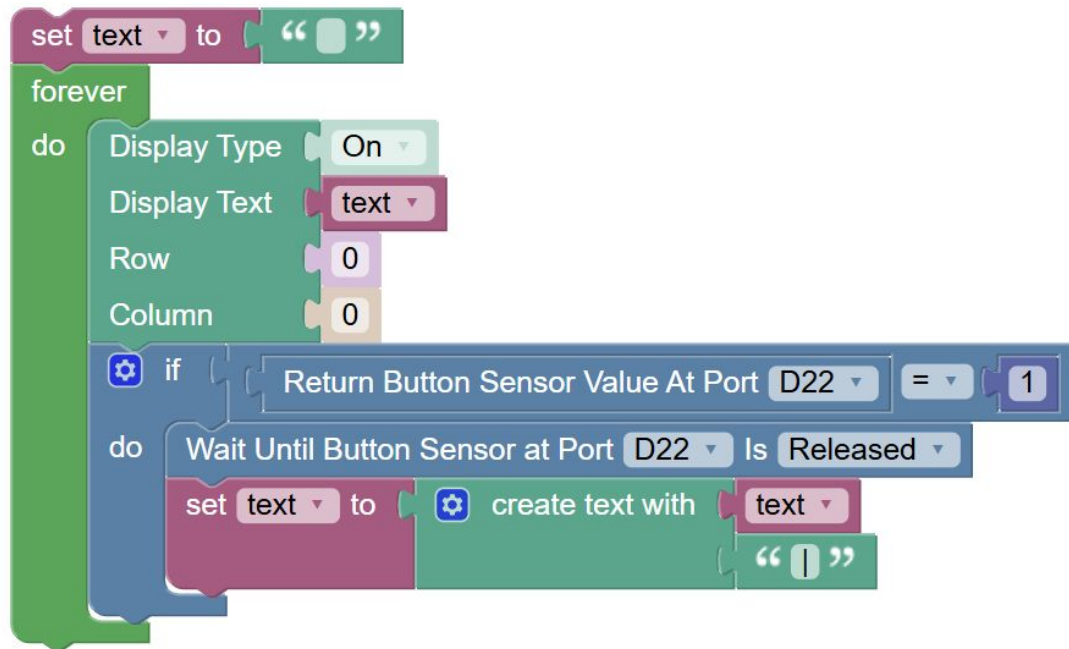
Create-text-with block appends bottom value to the top value.

Q2: D

Q3: C

The characters must be exact to be equal.

Q4:



3020 - Count Loop

Learning

A **Count loop** is a type of loop used to repeat actions while counting from one number to another. In most programming languages, this is commonly referred to as a **For loop**. It plays a key role in coding whenever sequencing or iterative manipulation is required.

In this unit, we will familiarise with how the Count loop function. You will see that it is an indispensable block when you learn about List in the next topic.

In class, we will explore the following:

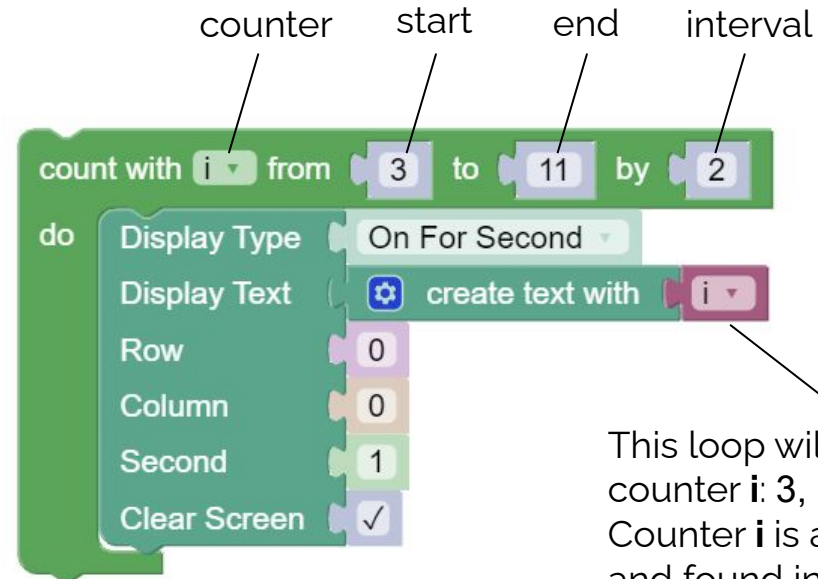
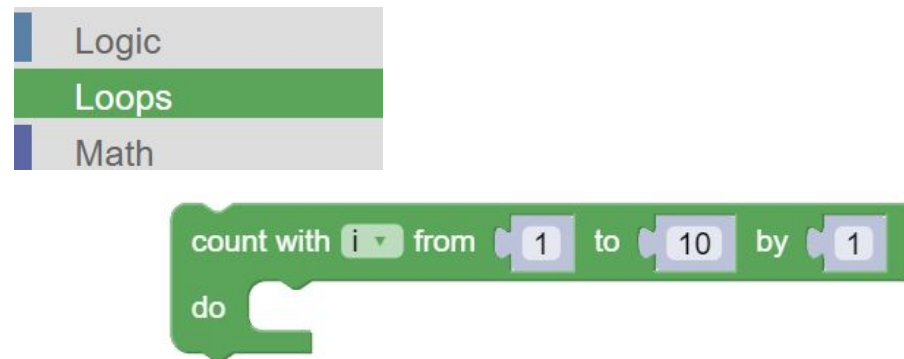
- Learn how to repeat within a range and generate number sequences.
- In computer science, clock logic serves as a classic model for understanding how numbers reset in cycles and how timed actions can interact or cascade. You will learn about clock logic through the development of a digital stopwatch.



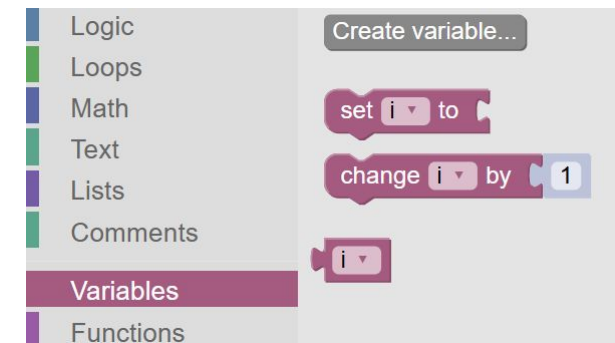
3020 - Count Loop

Count loop

Repeat codes within using a counting loop.

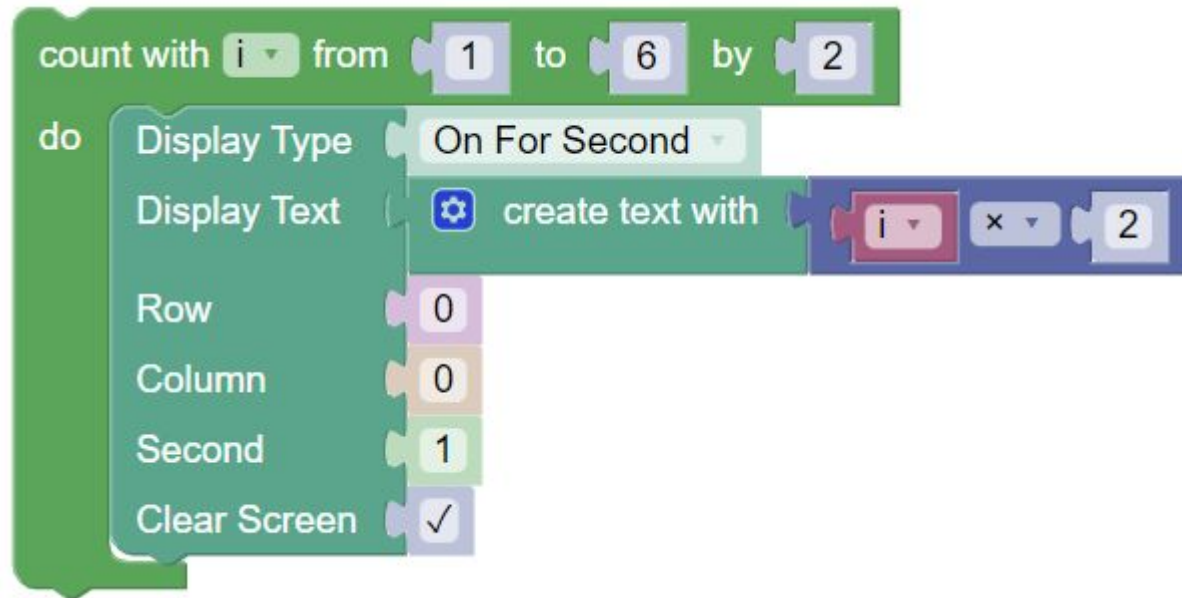


This loop will display the values of counter **i**: 3, 5, 7, 9, 11 (total 5 times). Counter **i** is a variable that is created and found in the variable section



Exercise Q1

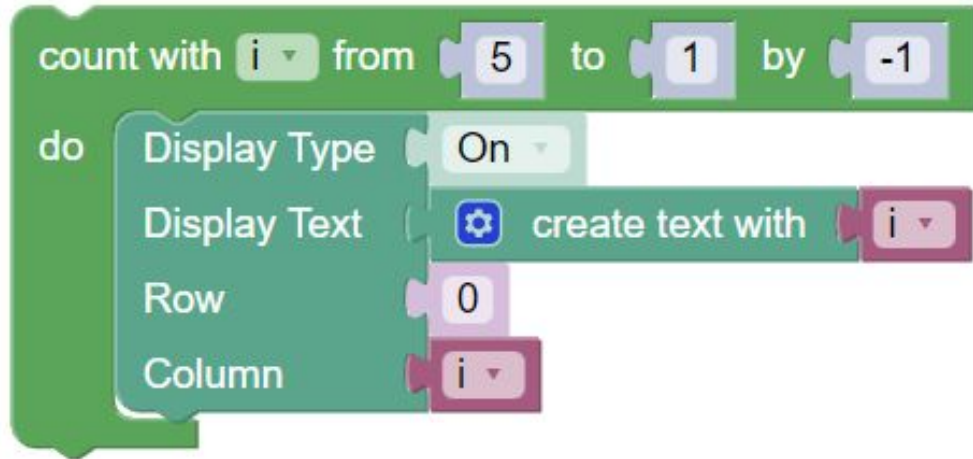
Which number sequence will be displayed on the LCD when the code is run?



- A) 1, 2, 3, 4, 5, 6
- B) 2, 4, 6
- C) 1, 3, 5
- D) 2, 6, 10

Exercise Q2

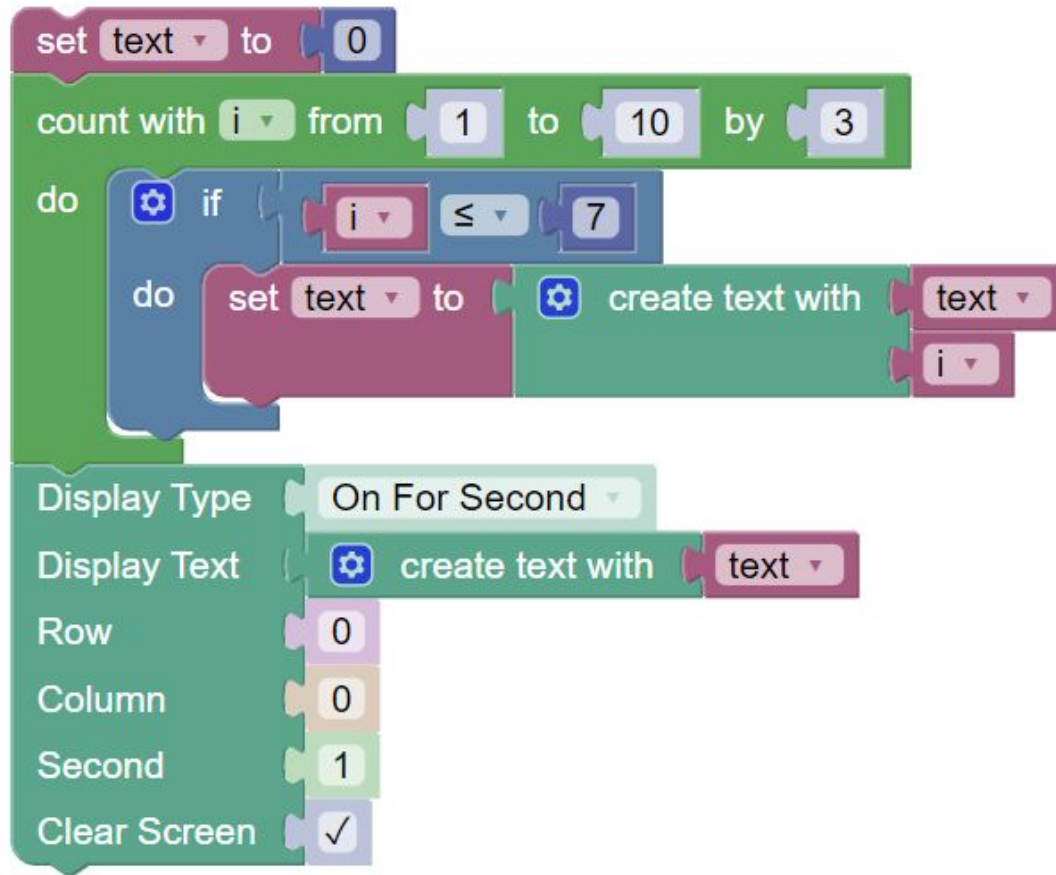
What will be displayed on the LCD after the following code has completed running?



- A) 12345
- B) 54321
- C) 51515
- D) iiii

Exercise Q3

What will be displayed on the LCD after the following code has completed running?



- A) 135
- B) 0136
- C) 0147
- D) 014

Exercise Q4

Create a code using count loop to produce the following animation.



Answers

Q1: D

Counter variable i will be 1, 3, 5 so multiply by 2 will give 2, 6, 10.

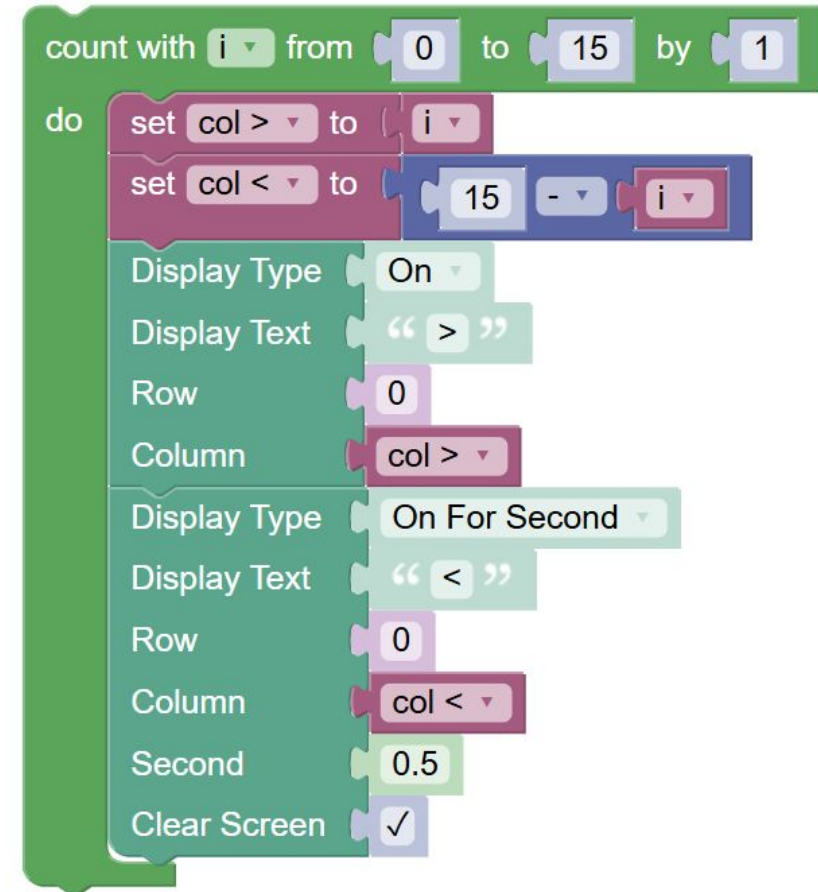
Q2: A

Counter variable i will be 5, 4, 3, 2, 1 and it will display at position 5, followed by 4 and so on. Hence, the display will look like 12345

Q3: C

Counter variable i will be 1, 4, 7, 10. So 1, 4, 7 will be appended to text, which begins with 0. Note that create-text-with block will convert all values to text.

Q4:

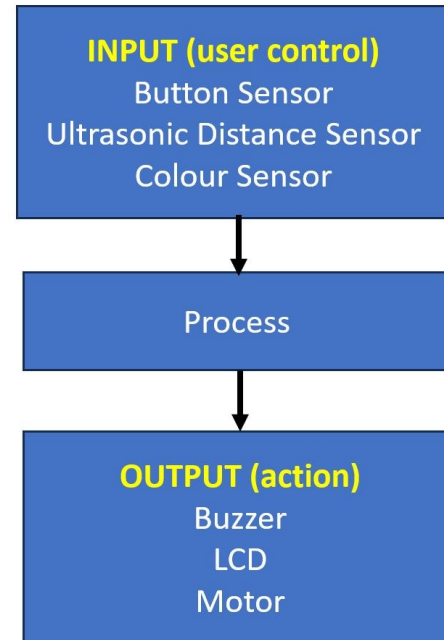


3021 - Intermediate 1 Project

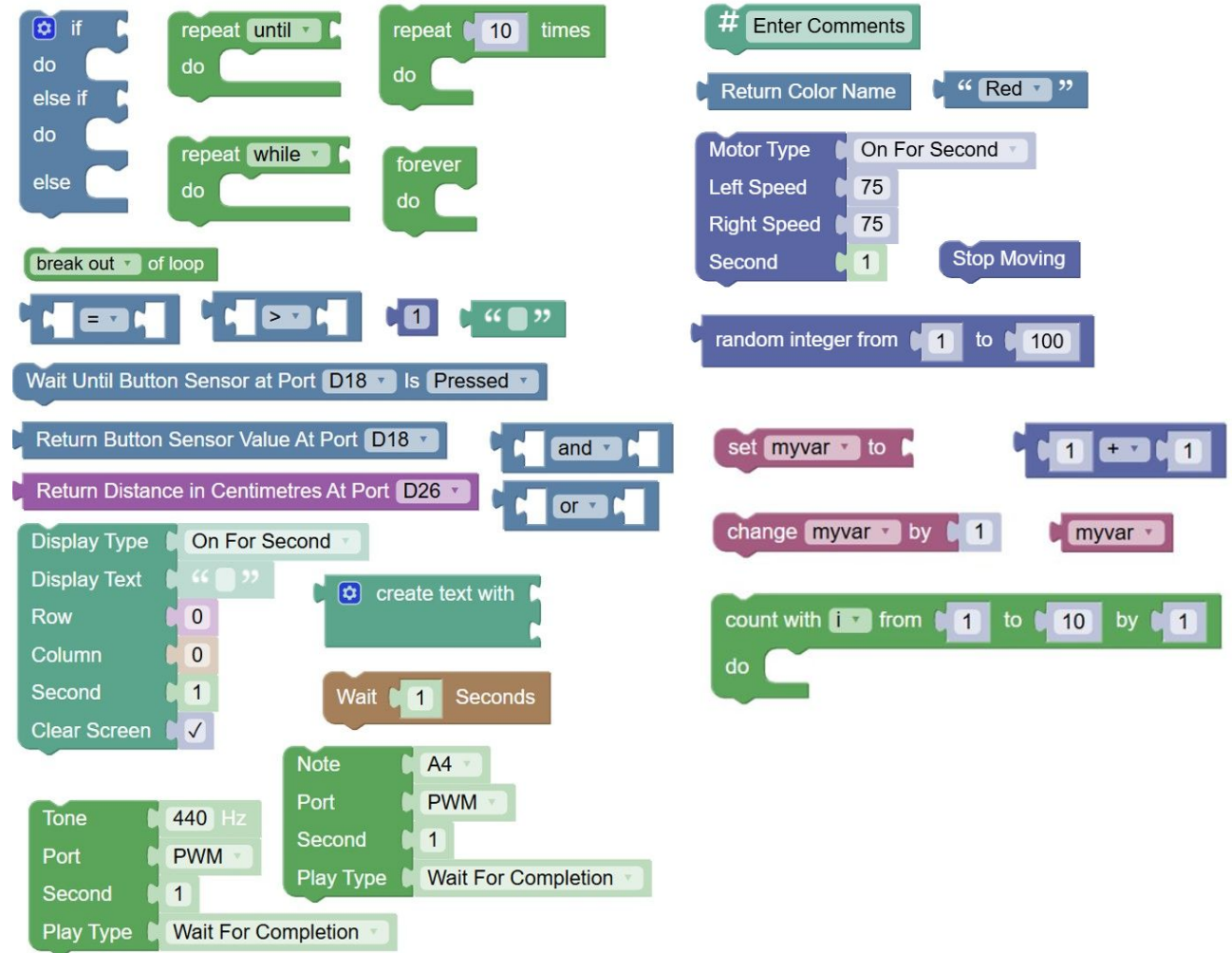
Learning

With a solid grasp of variables, your programs are no longer bound by rigid structures or tight constraints. Show how far your imagination can stretch. Let your next project be a playground for creativity and dynamic design!

Hardware learnt



Blocks learnt



Exercise Q1

With variables, you can now design a program with a lot more flexibility. Think of a project and dive in to make it happen!

To create a project, you need to:

1. Understand individual **block usage**
you can revise the notes and do the exercises to build a strong foundation
2. Practice **combining and applying** them to solve problems
you need to be attentive and diligent in class to master it
3. Think of a concept and **outline the sequence** of tasks
you can get creative or use an idea you came across before
4. Implement the outline **one part at a time**
all big problems are solved in small steps

3022 - List Basics

Learning

A **List** is used to store a group of values; each identified with a running index. In Blockly, this index starts from 1. The flexibility of List makes it a powerful tool for handling and organizing data in code.

In this unit, you'll learn how to create a list and read its contents.

In class, we will explore the following:

- Learn how to create and retrieve values from a list.
- Scrollable menu is a common application that is often implemented with a data structure like list. You will learn how to manipulate index to scroll and select a menu using list.



3022 - List Basics

List

A List is used to store a group of values; each identified with a running index starting from 1.
Think of it as a building (**list**) where each level (**index**) has a room that houses someone (**value**).

List	
index 1	value 1
index 2	value 2
index 3	value 3
index 4	value 4
index 5	value 5

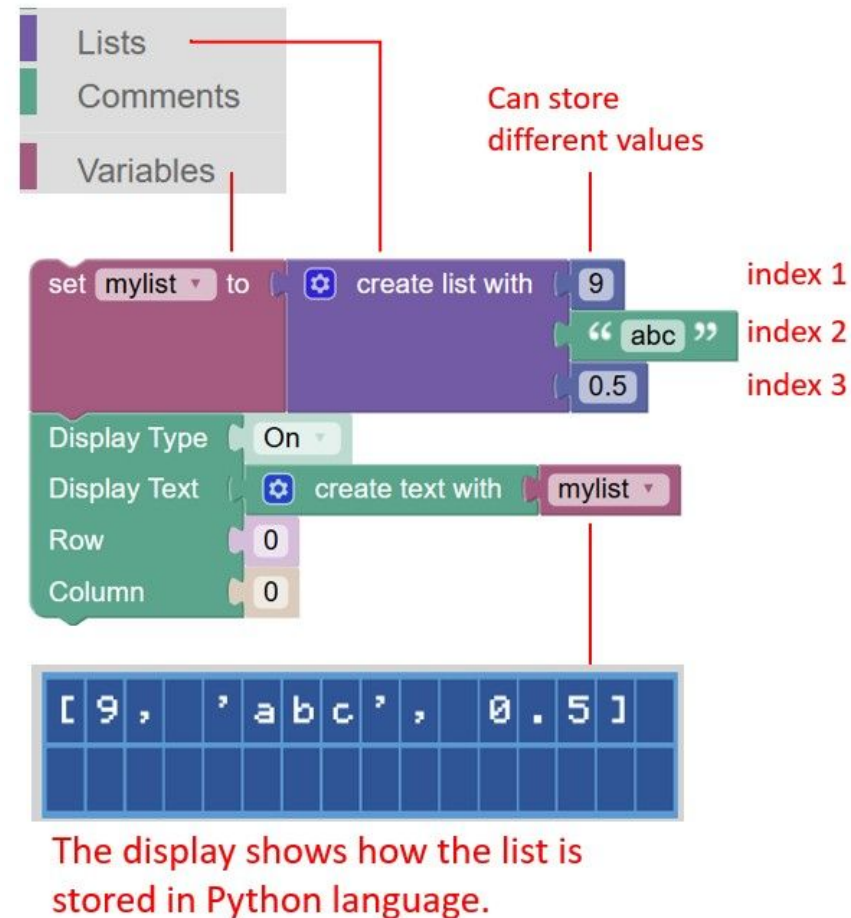
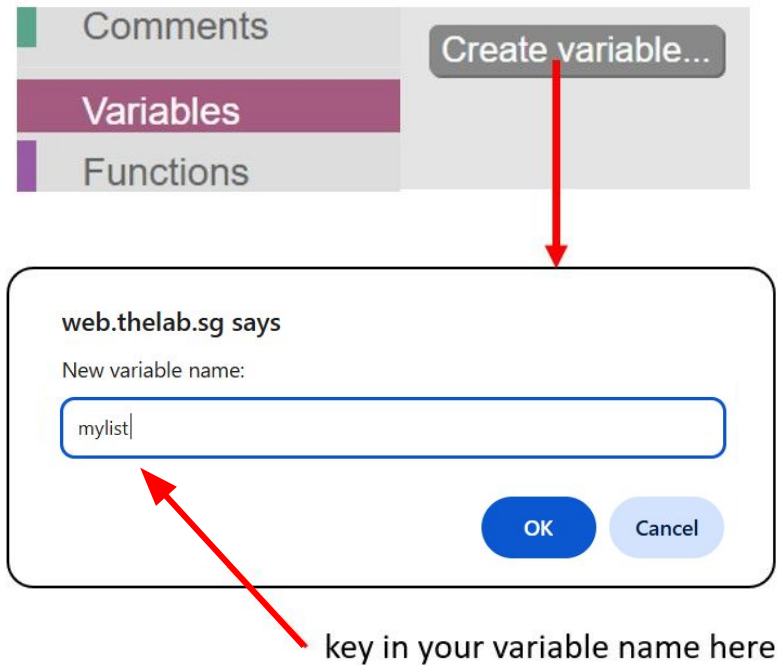
List[3] = value 3



3022 - List Basics

Creating a List

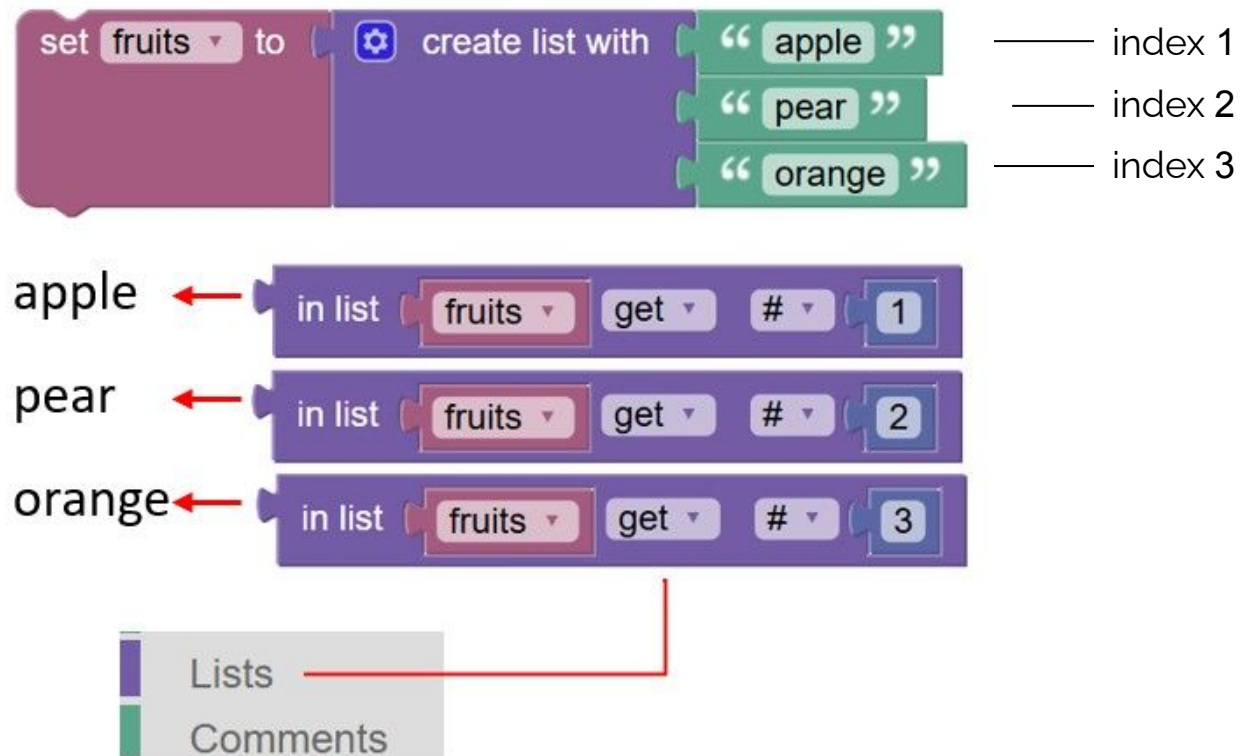
To create a list, first define a variable and give it a name. Then assign a list of values to that variable, as shown below.



3022 - List Basics

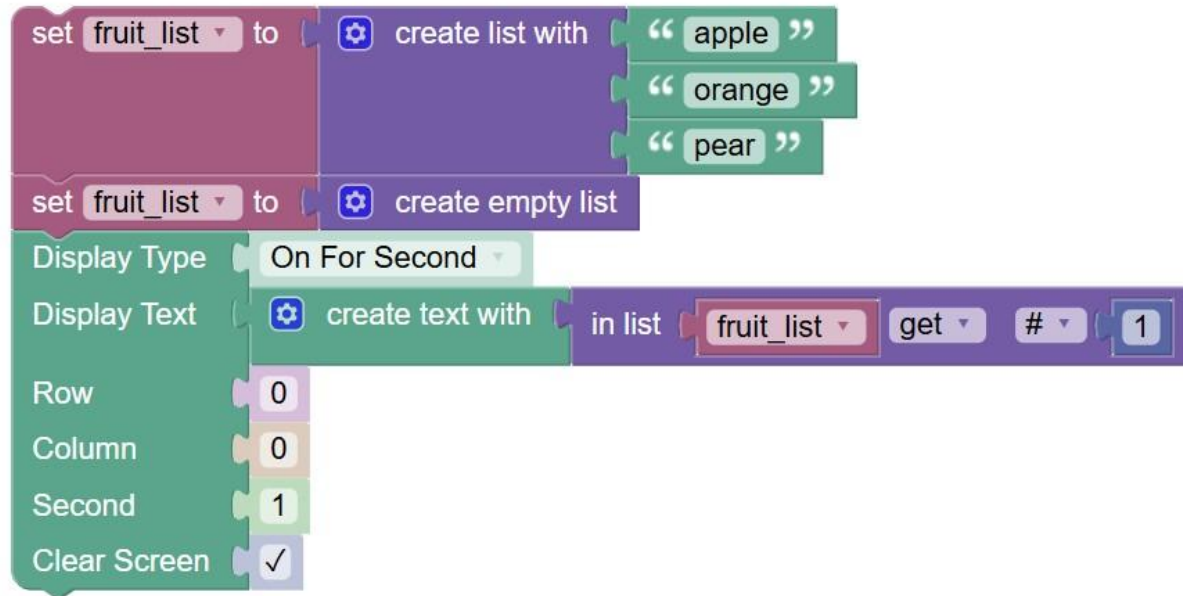
Reading a List

You can read each name by using the **List-Get block** as shown.



Exercise Q1

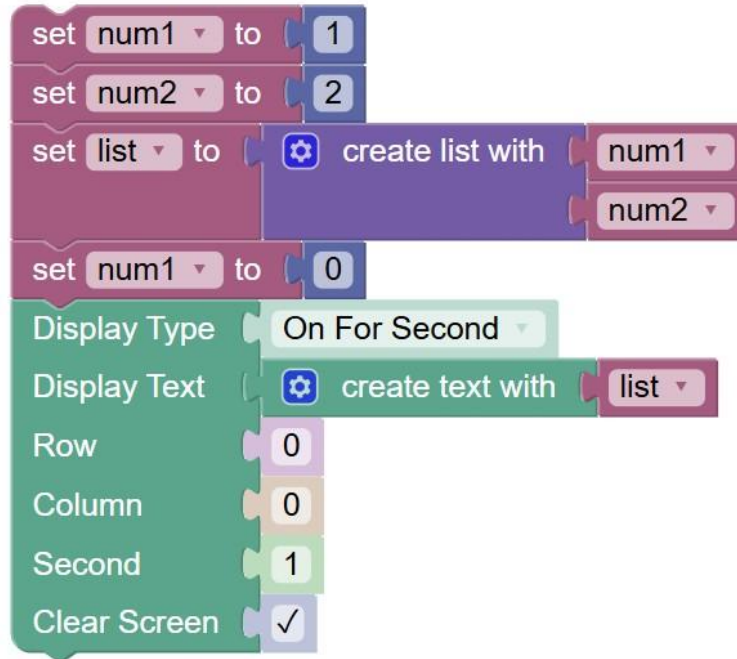
What will be shown on the LCD when the following code is run?



- A) apple
- B) pear
- C) None
- D) Nothing. The program will crash.

Exercise Q2

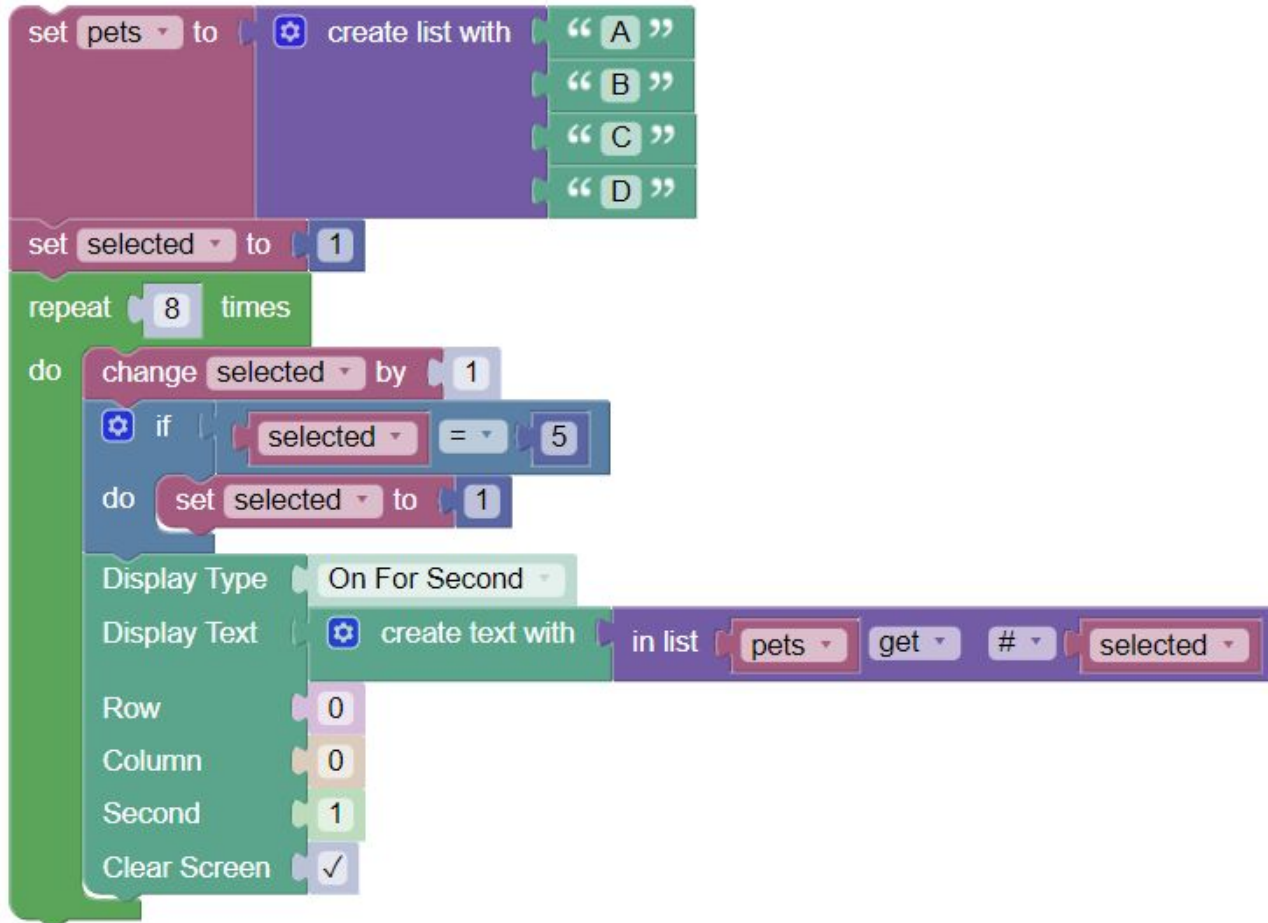
What will be shown on the LCD when the following code is run?



- A) [num1, num2]
- B) [12]
- C) [1, 2]
- D) [0, 2]

Exercise Q3

Which sequence will the following code display?



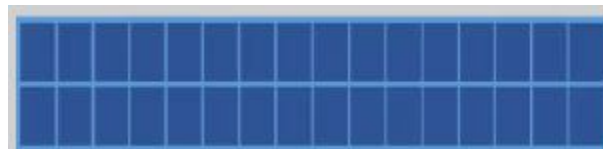
- A) 1, 2, 3, 4, 1, 2, 3, 4.
- B) A, B, C, D, A, B, C, D.
- C) B, C, D, A, B, C, D, A.
- D) A, A, A, A, A, A, A, A.

Exercise Q4

Below is a list of fishes and its quantity.



Use a loop to display the information one at a time as shown.



Answers

Q1: D

fruit_list is set to empty and retrieving value at index 1 is invalid.

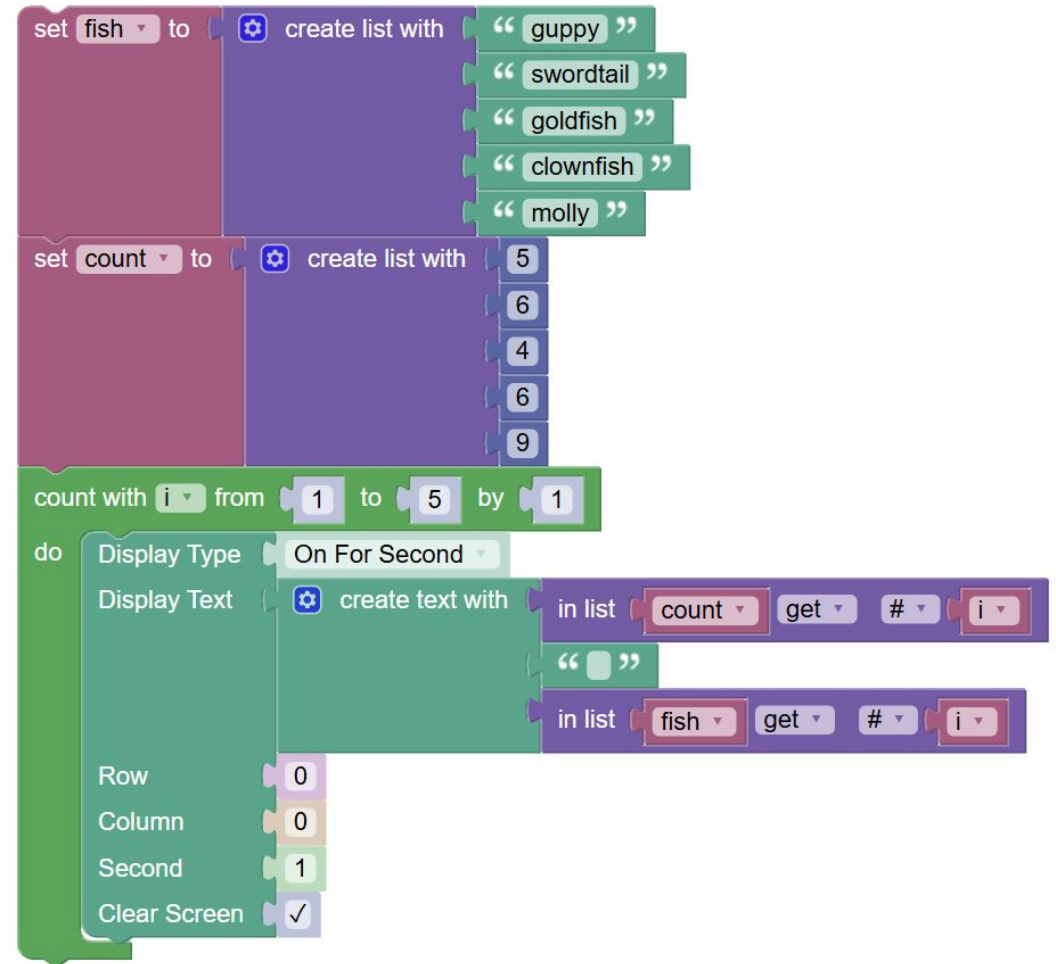
Q2: C

The list stores the initial values of num1 and num2, which is 1 and 2.

Q3: C

The variable selected is used as the index. Based on the sequence, it will start by displaying value at index 2, then 3, 4, and resets to 1. It repeats again 2, 3, 4, 1. So the displayed values will be B, C, D, A, B, C, D, A.

Q4:



3023 - List Updating

Learning

You learnt how to create and read values from a List in the previous unit.

In this unit, you will learn how to loop through a list (iterate) and modify its values.

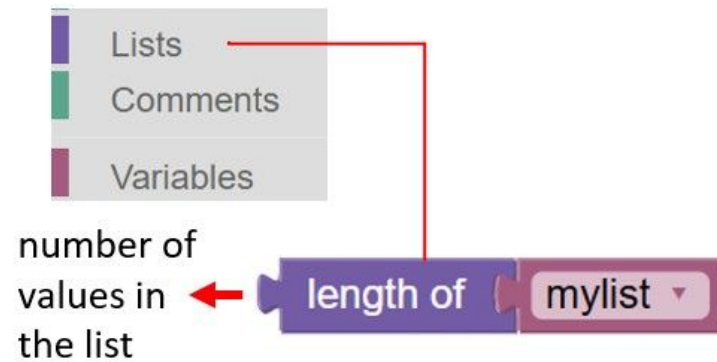
In class, we will explore the following:

- Familiarising with list iteration and value modification.

3023 - List Updating

List Length

The **Length block** returns the number of items (values) in a list.



3023 - List Updating

Iterating through a List

Iterating through a list means to go through a list and access every value one by one.

There are different ways to iterate through a list. One way is using the **For-each loop** as shown below.

Each value in the list is copied to variable **item** as it loops.

The loop will stop when it reaches the last value.



the default variable is **i** but you can use any variable.

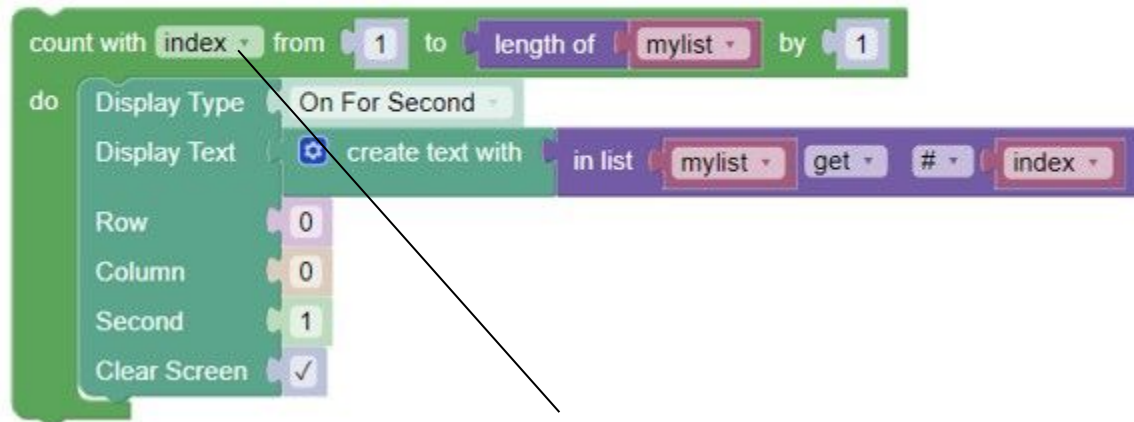
3023 - List Updating

Iterating through a List

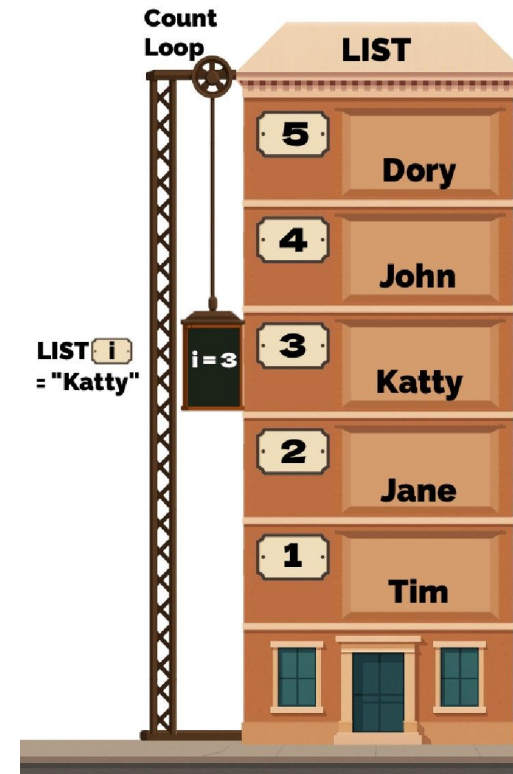
Another way is using a **Count loop**.

We visualised a list as a building. Now, imagine a lift beside it, stopping at each floor, just like a Count-loop counting from 1 to the top. As the lift ascends, it stores the current floor number in the counter variable, which acts as the index to access values in the list.

The variable **index** is a counter that counts from 1 to the last position. Use **Get block** to read the value at index. This method can control the range of items to read.



the default variable is **i** but you can use any variable.



3023 - List Updating

Modifying value in a List

The **List-Set block** is used to change a value in a list.

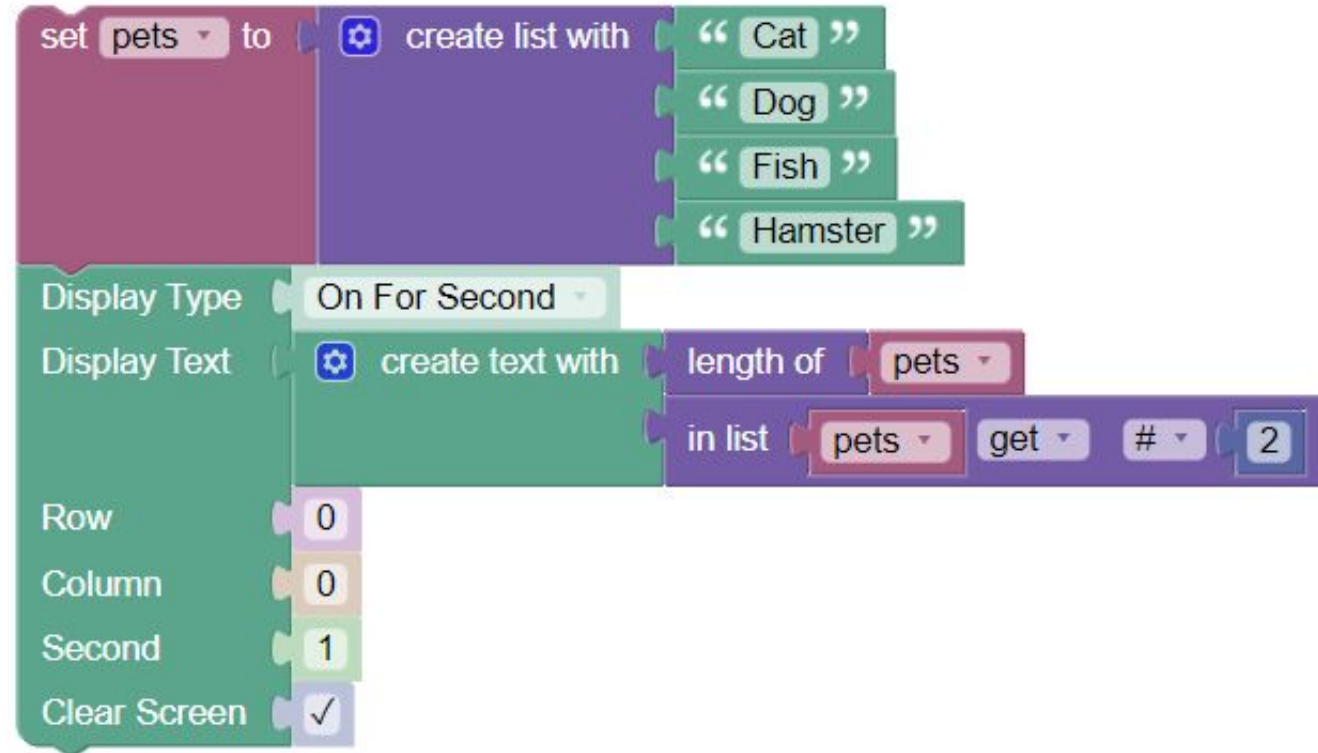
Changing the value of **mylist** at **index 3** to **“new value”**



Exercise Q1

What will be displayed on the screen when the code is run?

- A) 2Dog
- B) 3Dog
- C) 4Dog
- D) DogDog



Exercise Q2

Which of the following can randomly select a value from the list?

- A) Diagram 1 only
- B) Diagram 2 only
- C) Diagram 3 only
- D) Diagrams 1, 2 and 3

Diagram 1

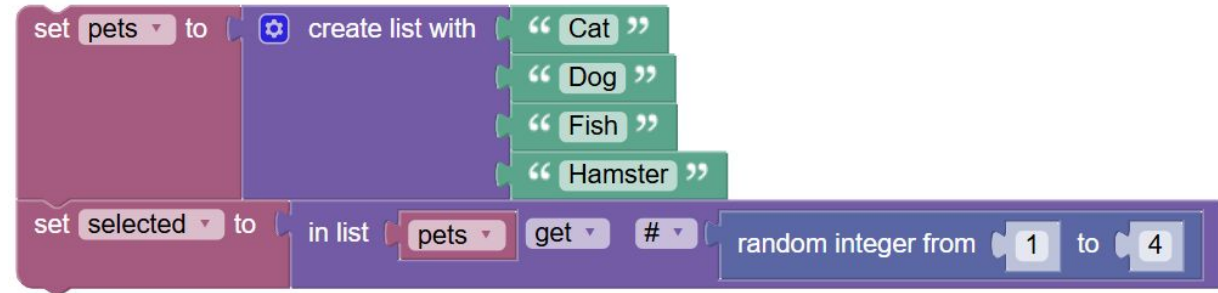


Diagram 2

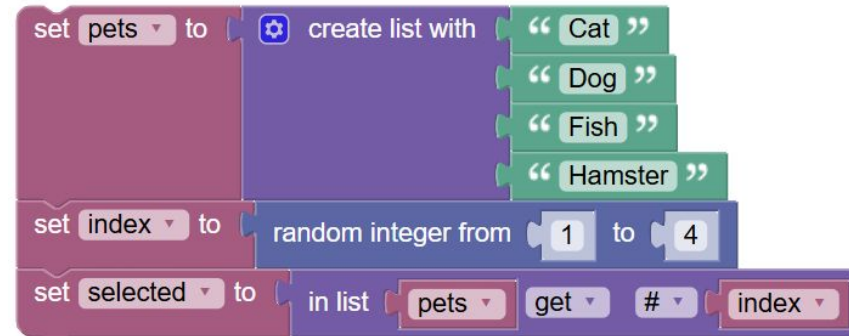
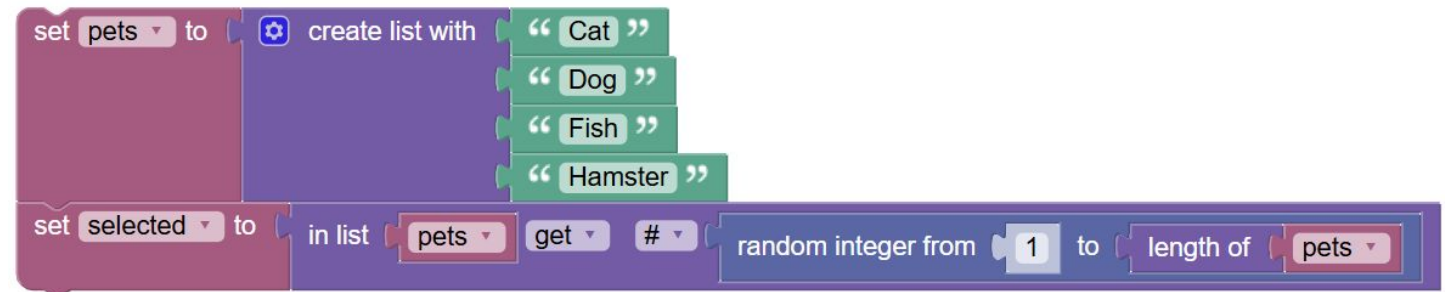


Diagram 3



Exercise Q3

In which of the following will pets list become ['Cat', 'Dog', 'Terrapin', 'Hamster']?

- A) Diagram 1 only
- B) Diagram 2 only
- C) Diagram 3 only
- D) Diagrams 2 and 3 only

Diagram 1



Diagram 2

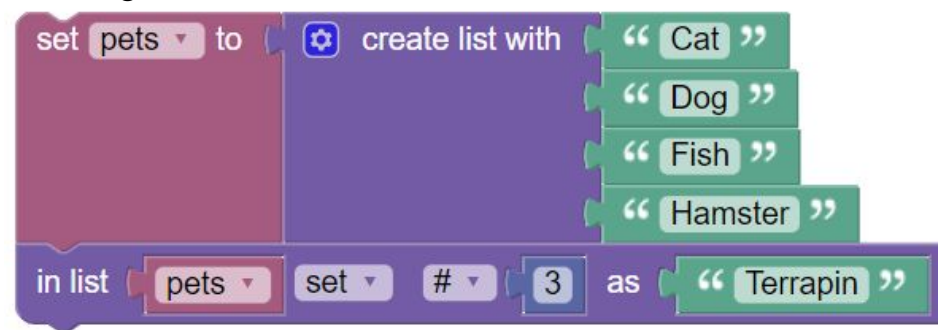


Diagram 3



Exercise Q4

Which of the following can display all the values in the list one by one?

- A) Diagrams 1 and 2 only
- B) Diagrams 1 and 4 only
- C) Diagrams 2 and 3 only
- D) Diagrams 3 and 4 only

Diagram 1



Diagram 3

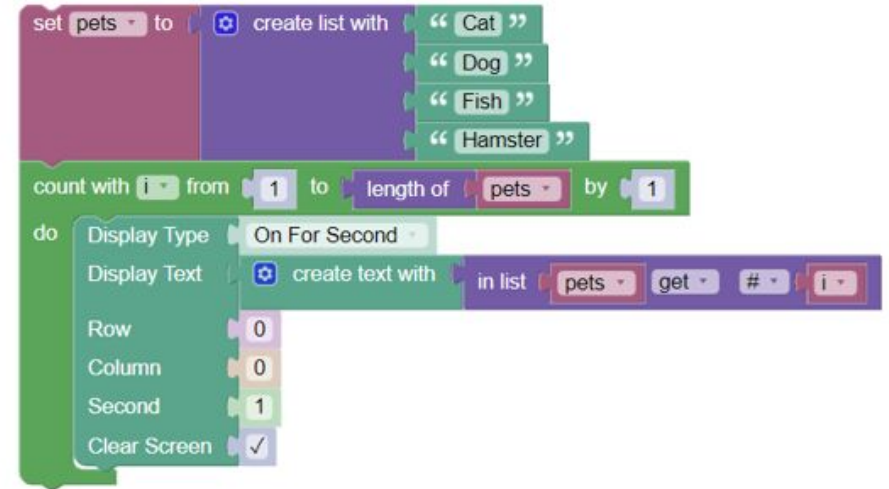


Diagram 2

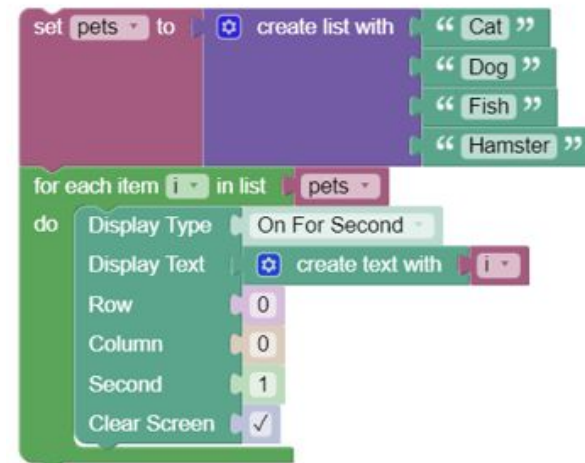
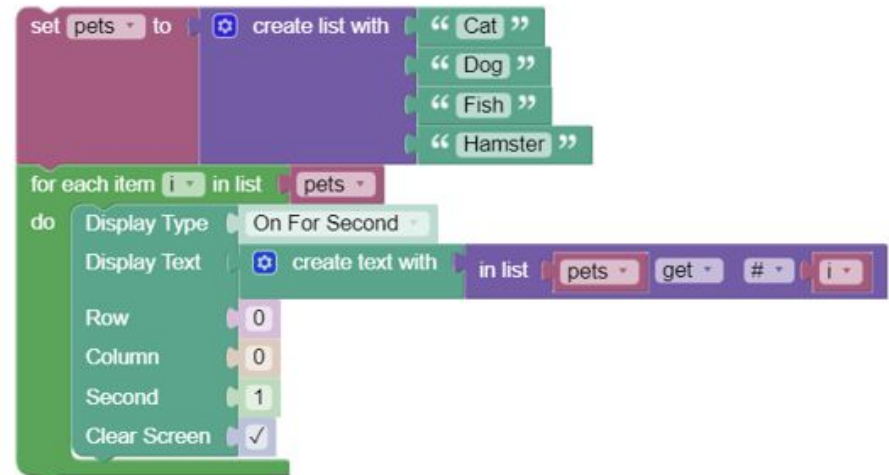


Diagram 4



Answers

Q1: C

Length of pets list is 4. Value at item 2 is Dog.
create-text-with block will join 4 and "Dog" to give "4Dog"

Q2: D

As long as the selection range is valid (1 to 4), it will work.

Q3: B

The list will become as follow:

Diagram 1, 'Terrapin'

Diagram 2, ['Cat', 'Dog', 'Terrapin', 'Hamster']

Diagram 3, ['Cat', 'Dog', 'Terrapin', 'Fish', 'Hamster']

Q4: C

Using count loop, i is the index so you need to use the get block to retrieve the value. Using "for each loop" block, i stores the value in the list instead, so you can display it directly.

Diagram 1 displays 1,2,3,4.

Diagram 4 has error as i is not an index but the list values.

3024 - List Manipulation

Learning

A List can expand or shrink during the program execution. This is what makes it ideal for managing changing data in real-time.

In this unit, you will learn how to add and remove items from a list.

In class, we will explore the following:

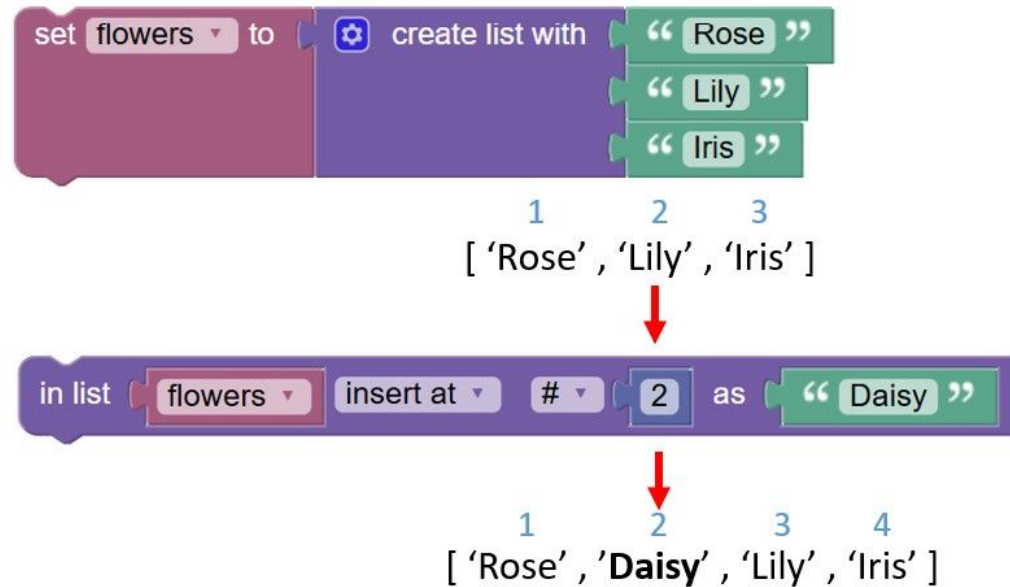
- Adding and removing items from a list while setting lower and upper limits.
- Applications including a fun Memory game with unpredictable outcome.



3024 - List Manipulation

Inserting value into of List

You can insert a value at any index in a list as shown:



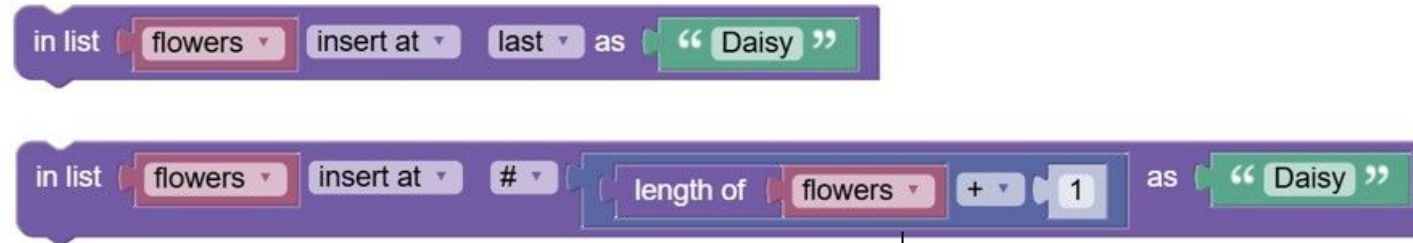
Note that the list must exist (can be empty or created with initial values) before inserting.



3024 - List Manipulation

Adding Value to the End of List

You can insert a value to the end of a list (append) as shown:

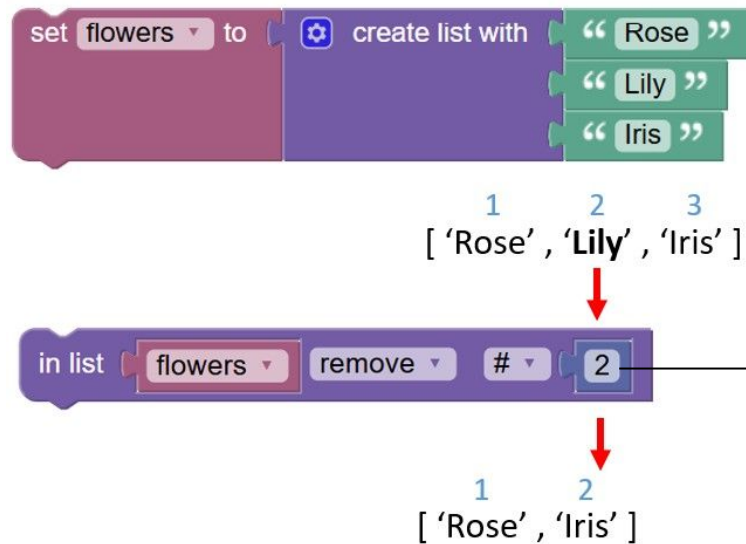


e.g. if there are 4 items in the list,
insert at index 5

3024 - List Manipulation

Removing value from List

You can remove any value from a list.



Indicate the index of the value to remove

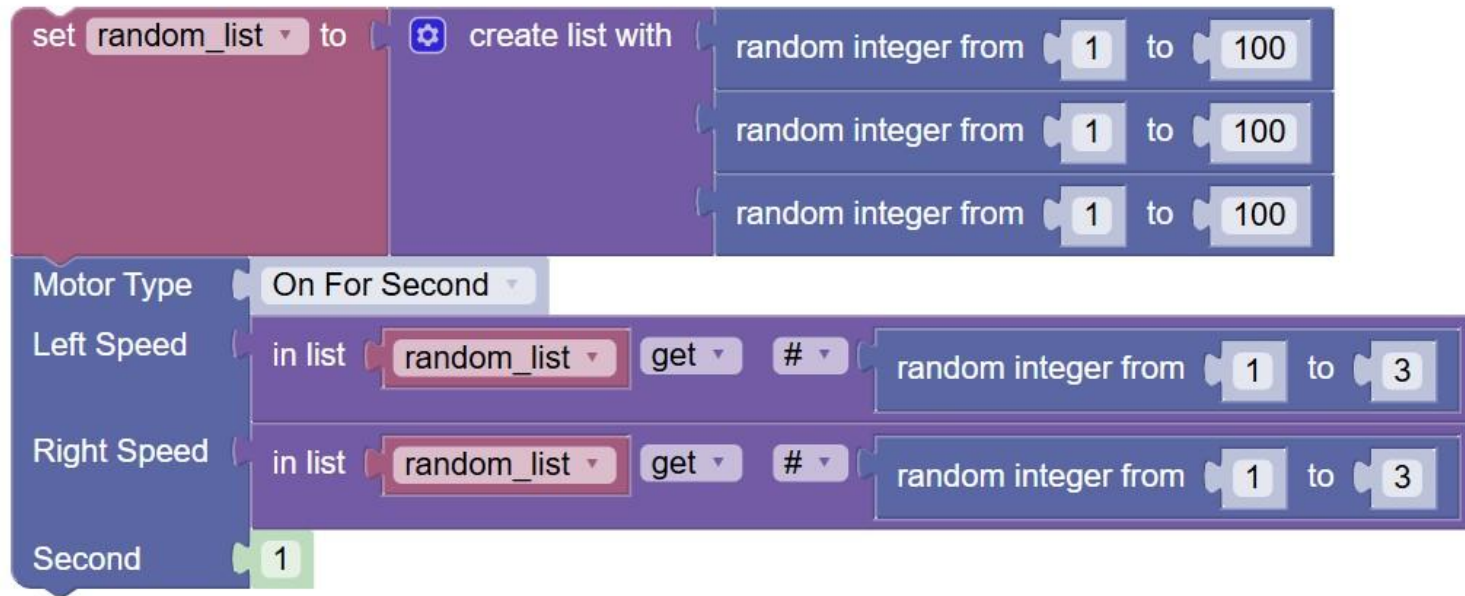
or delete all its values as shown.



Note that attempting to remove a value from an invalid index (or empty list) will crash the program.

Exercise Q1

Which statement is correct about the code snippet below?

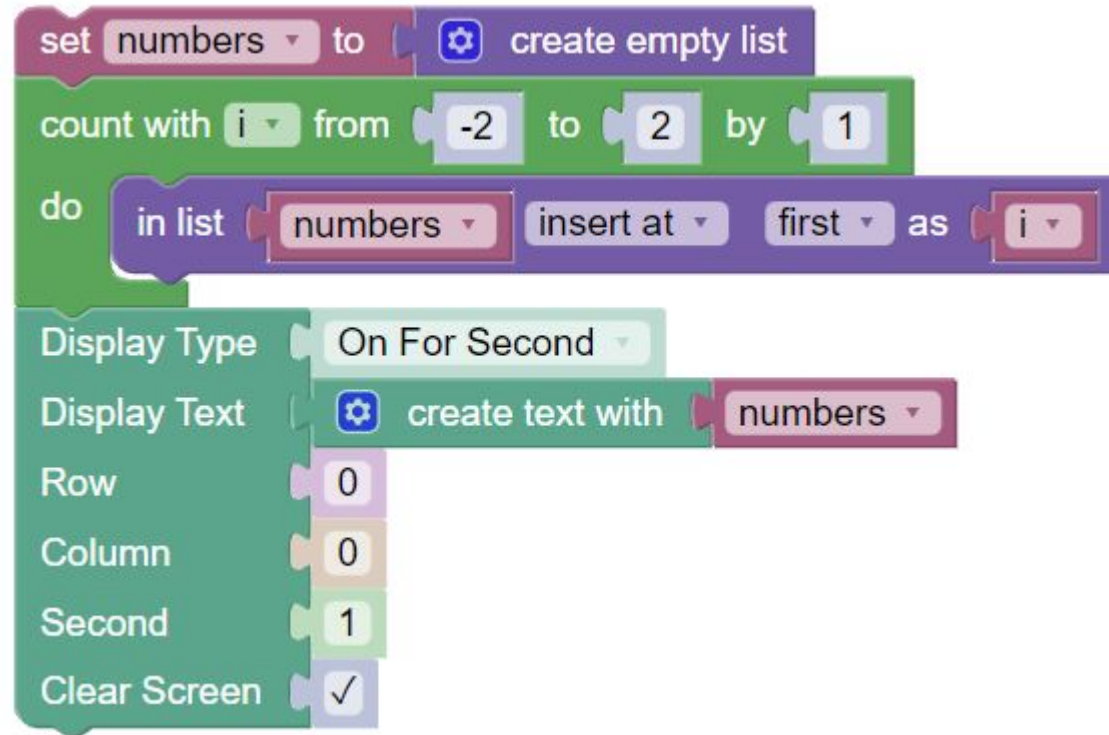


- A) All the three values in the random list will always be the same.
- B) The left and right speeds of the motor will always be the same.
- C) The program will crash.
- D) None of the above.

Exercise Q2

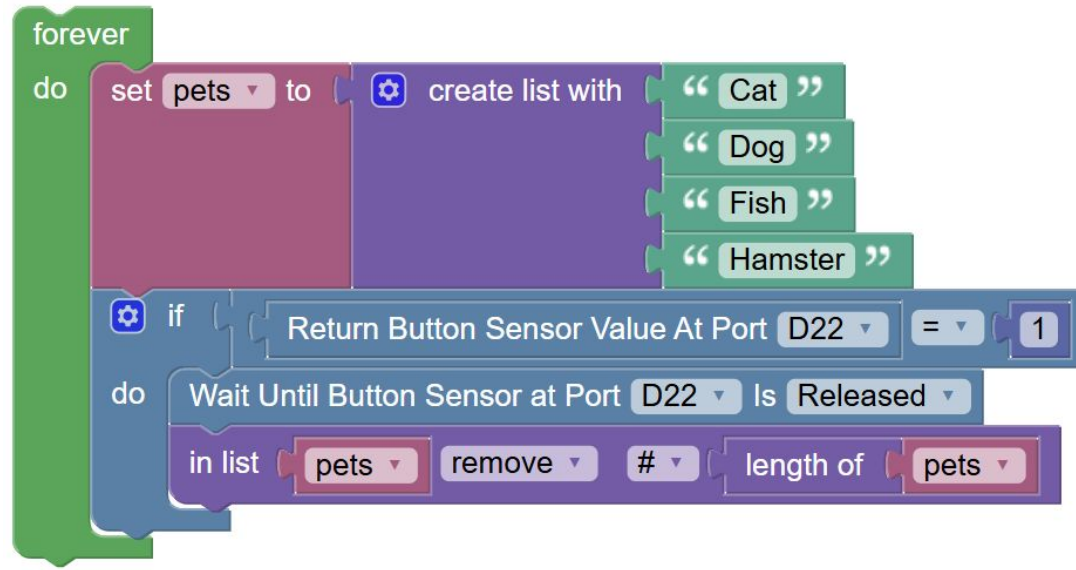
What will be displayed on the screen when the code is run?

- A) []
- B) [-2, -1, 2]
- C) [-2, -1, 0, 1, 2]
- D) [2, 1, 0, -1, -2]



Exercise Q3

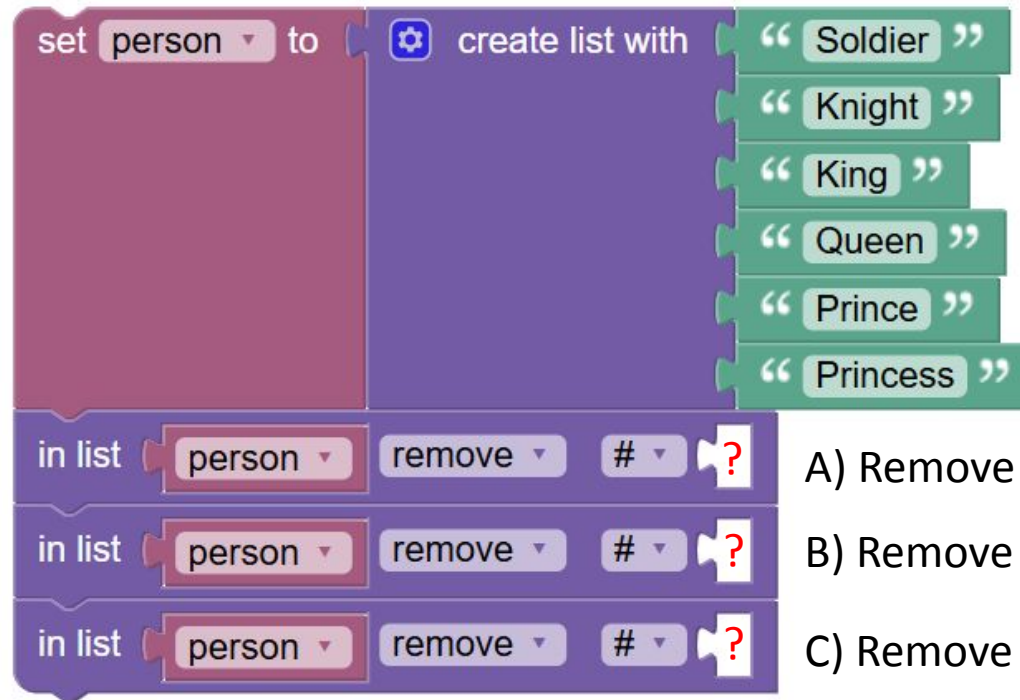
Which of the statements correctly described what happen when the button is **pressed and released repeatedly**?



- A) The values in the pets list will be deleted one by one from the last to the first until it is empty. The program will continue and nothing happens when the button is pressed and released.
- B) The values in the pets list will be deleted one by one until the list is empty. After that, the program will crash if it attempts to delete from the empty list.
- C) Only 'Hamster' will be deleted no matter how many times the button is pressed.
- D) Only 'Cat' will be deleted no matter how many times the button is pressed.

Exercise Q4

Write down what should be the index number in the code below to remove Soldier, King and Princess in order.



The code consists of the following blocks:

- set person to** (purple block)
- create list with** (purple block) with the following items:
 - “ Soldier ”
 - “ Knight ”
 - “ King ”
 - “ Queen ”
 - “ Prince ”
 - “ Princess ”
- in list** (purple block) with **person** selected, **remove** selected, and a question mark in the index field.
- in list** (purple block) with **person** selected, **remove** selected, and a question mark in the index field.
- in list** (purple block) with **person** selected, **remove** selected, and a question mark in the index field.

To the right of the code blocks are three options:

- A) Remove Soldier
- B) Remove King
- C) Remove Princess

Answers

Q1: D

Each time the random block is used, a new random number will be generated. The list will contain 3 random numbers though they may repeat. The left and right speeds will be assigned a random choice from the list.

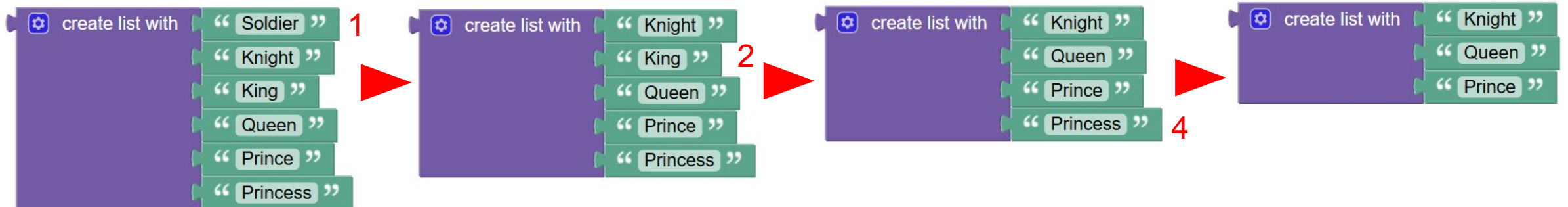
Q2: D

Counter variable i will be -2, -1, 0, 1, 2. When inserted at the front each time, the order will appear reversed 2, 1, 0, -1, -2.

Q3: C

In the loop, the pets list is reset to the 4 values every time. So the last value, Hamster, is always removed each round.

Q4: A) 1 B) 2 C) 4



3025 - Function

Learning

Programs are often split into smaller tasks, each written as a function. Functions make code easier to read, reuse, and organize, especially when the same instructions are needed again and again.

In this unit, you'll learn how to organize code into functions.

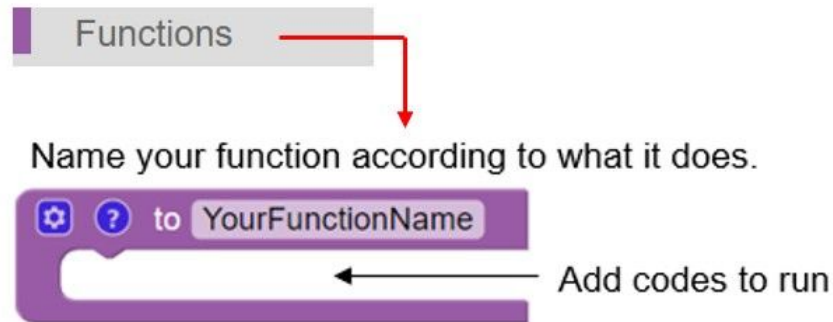
In class, we will explore the following:

- Creating simple functions.
- Organising your previous program into functions to understand the flow of function calls.

3025 - Function

Function

Use the **Function block** to create a function. Name you function meaningfully (usually the task it performs) and add in the codes to be run. Place this anywhere in the codin space. It will not run until it is called.



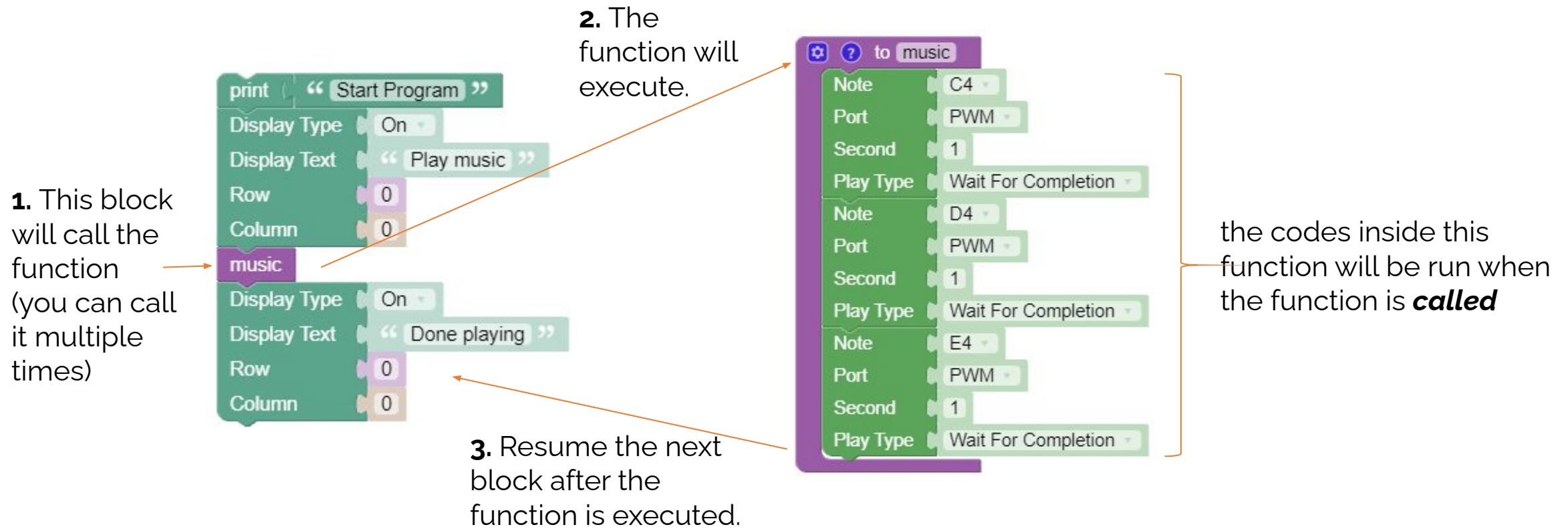
A block with the function name will appear once the function is created. Use it to run the function.



3025 - Function

Function

Below is an example of the execution.



Exercise Q1

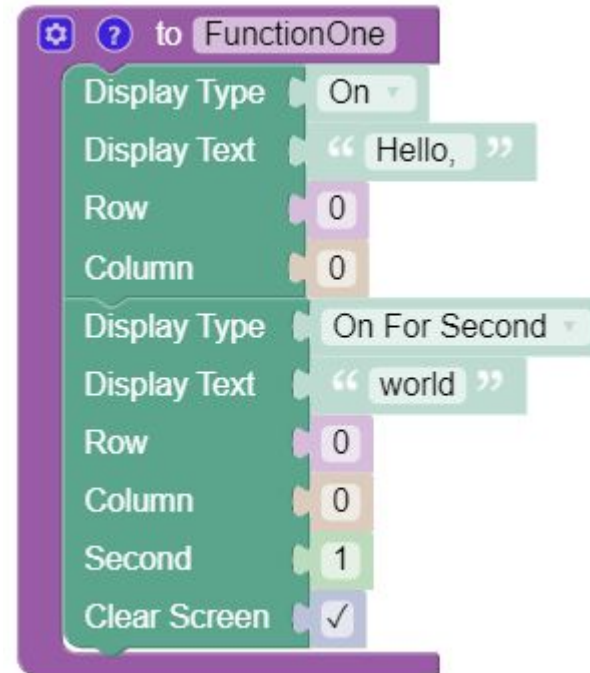
Which of the following is true about Function?

- A) Functions are useful for designing a program as a set of smaller tasks.
- B) Functions help reduce repeated code segments.
- C) Functions enable effective team programming.
- D) All of the above.

Exercise Q2

A program only has the following code. When the program is run, how many times will "Hello, world" be displayed?

- A) 0
- B) 1
- C) 2
- D) It will be displayed repeatedly



Exercise Q3

When the following code is run, how many times will "Hello, world" be displayed?

- A) 0
- B) 1
- C) 2
- D) 3

FunctionOne
FunctionTwo
FunctionOne

to FunctionOne

- Display Type: On
- Display Text: "Hello, "
- Row: 0
- Column: 0
- Display Type: On For Second
- Display Text: "world "
- Row: 0
- Column: 7
- Second: 1
- Clear Screen: ☒

to FunctionTwo

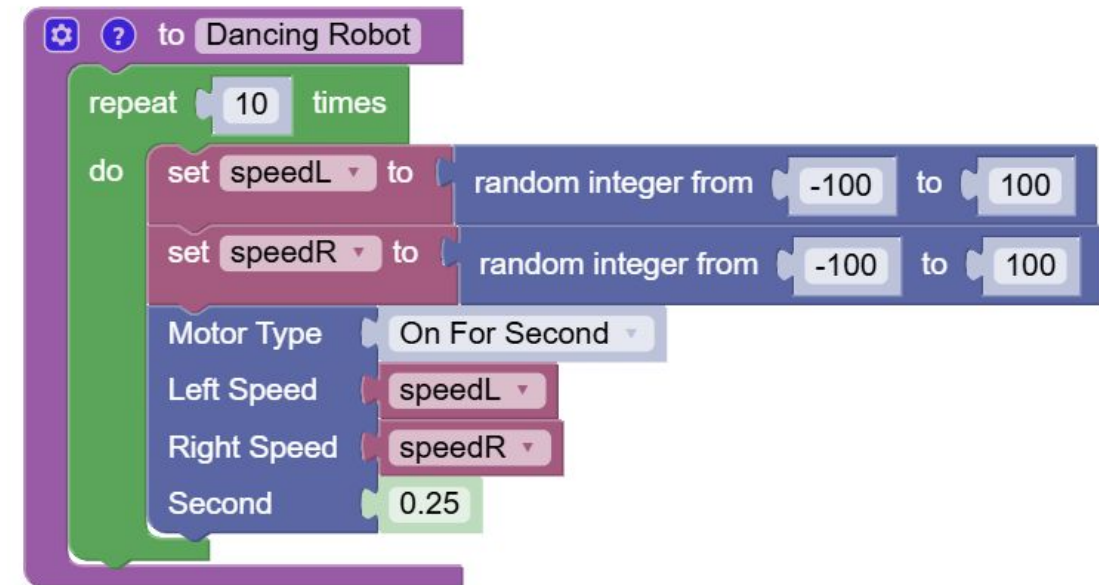
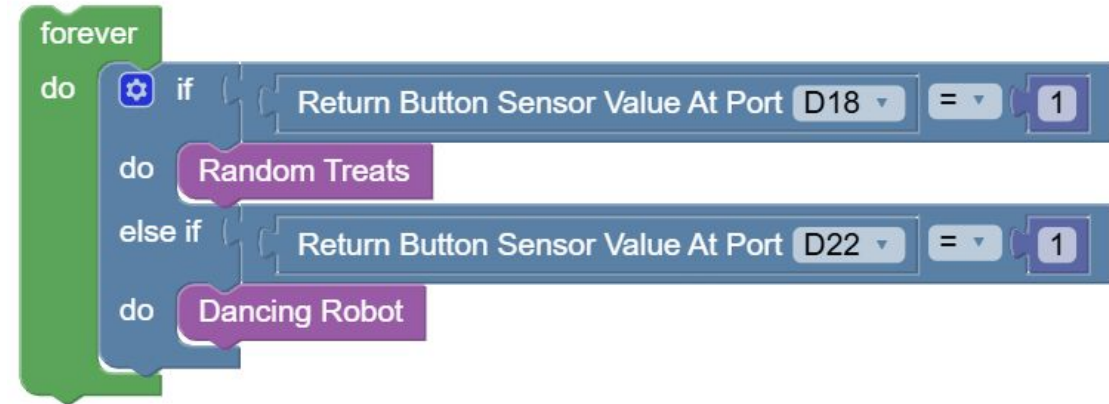
- Note: A4
- Port: PWM
- Second: 1
- Play Type: Wait For Completion
- Note: B4
- Port: PWM
- Second: 1
- Play Type: Wait For Completion

Exercise Q4

In the program below, implement the function “Random Treats” to randomly select a treat from a list and display it.

The treats consist of:

- Lollipop
- Ice-cream
- Chocolate
- Cupcake



Answers

Q1: D

Q2: A

There is no function call so the function will not be run.

Q3: C

FunctionOne is called twice so it will display "Hello, world" twice.

Q4:



3026 - Function Parameter

Learning

A function typically requires input values to perform its task and produce a result. These input values are stored in **parameters**, which act as storage for the data passed into the function during its execution.

In this unit, you will learn about function with parameters.

In class, we will explore the following:

- Creating function with parameters.
- Applications including a digital billboard that displays animated display.

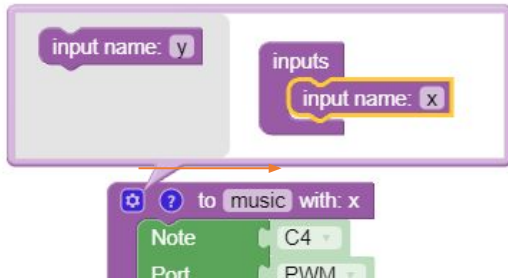


3026 - Function Parameter

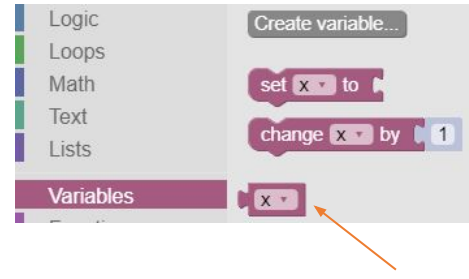
Function with parameters

Function can take in values to process. The values passed to a function are called **arguments** and the variables that hold them in the function are called **parameters** in computing terms.

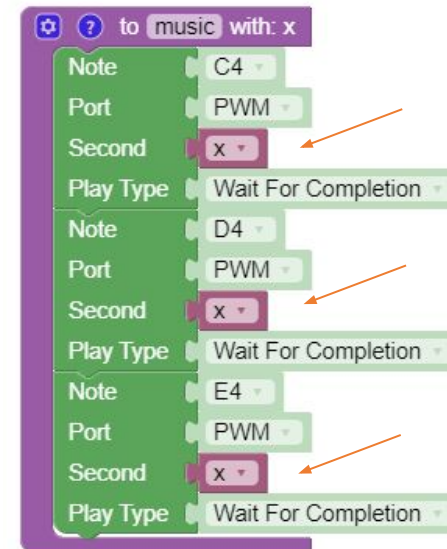
You can add any number of parameters as shown.



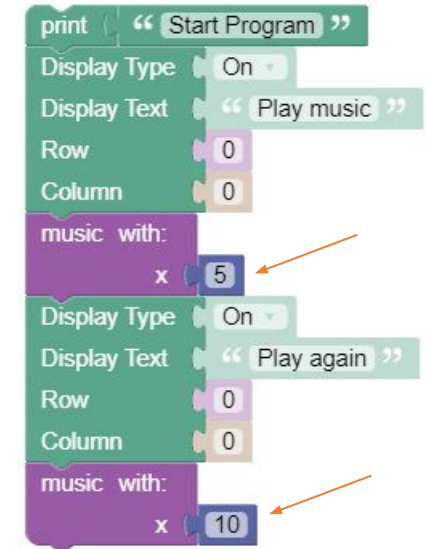
The parameters will appear in the variables menu.



The example below uses the parameter value to control the duration of sound.



Call the function by supplying it with a value (argument) to the parameter.

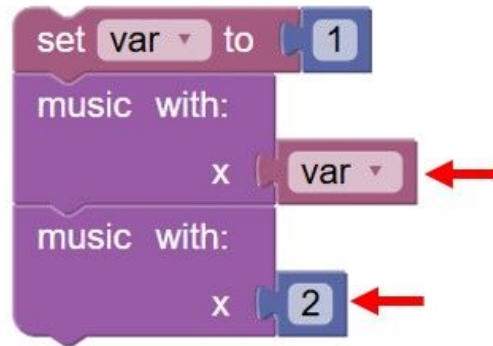


3026 - Function Parameter

Function with parameters

The argument can be the value itself or a variable. However, make sure the value passed to a function is appropriate.

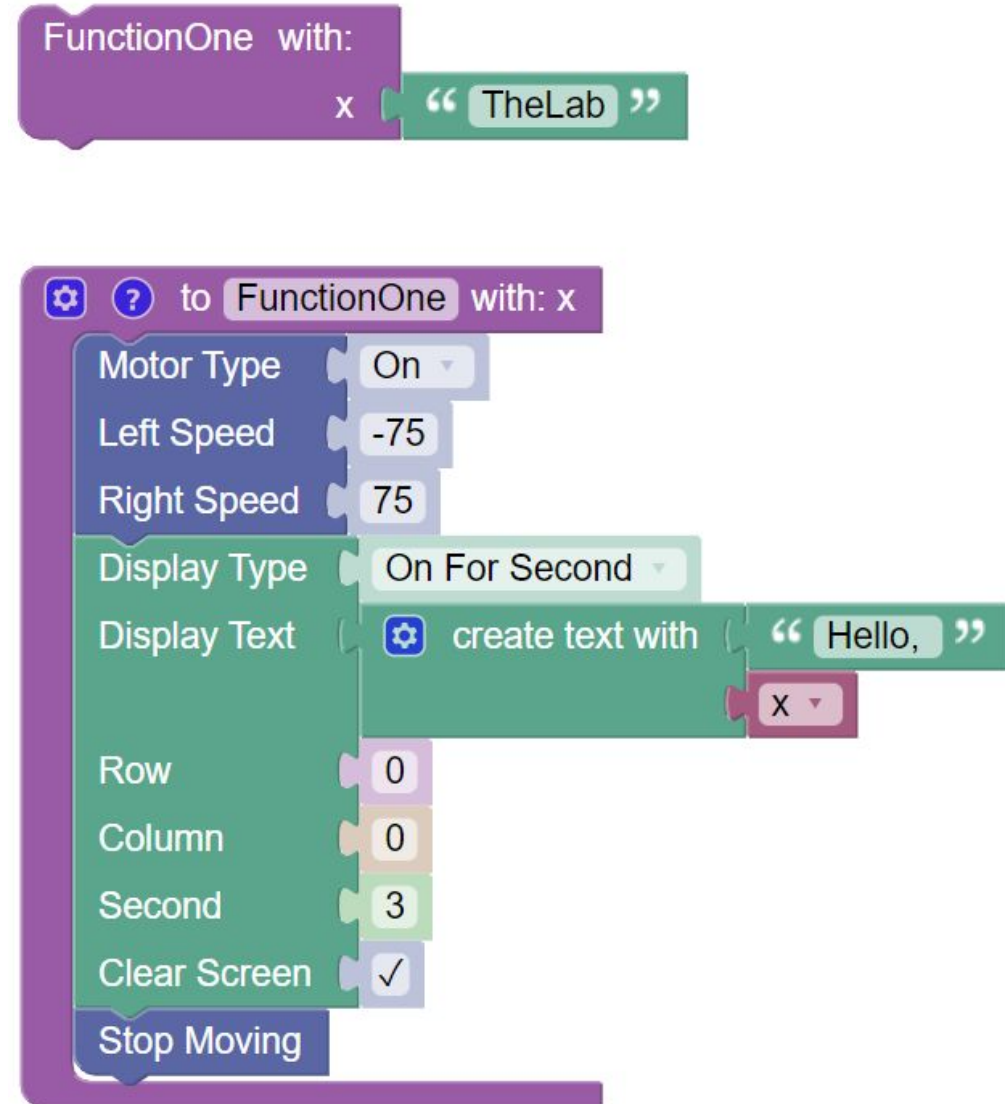
You can pass value to the parameter directly or through a variable.



Exercise Q1

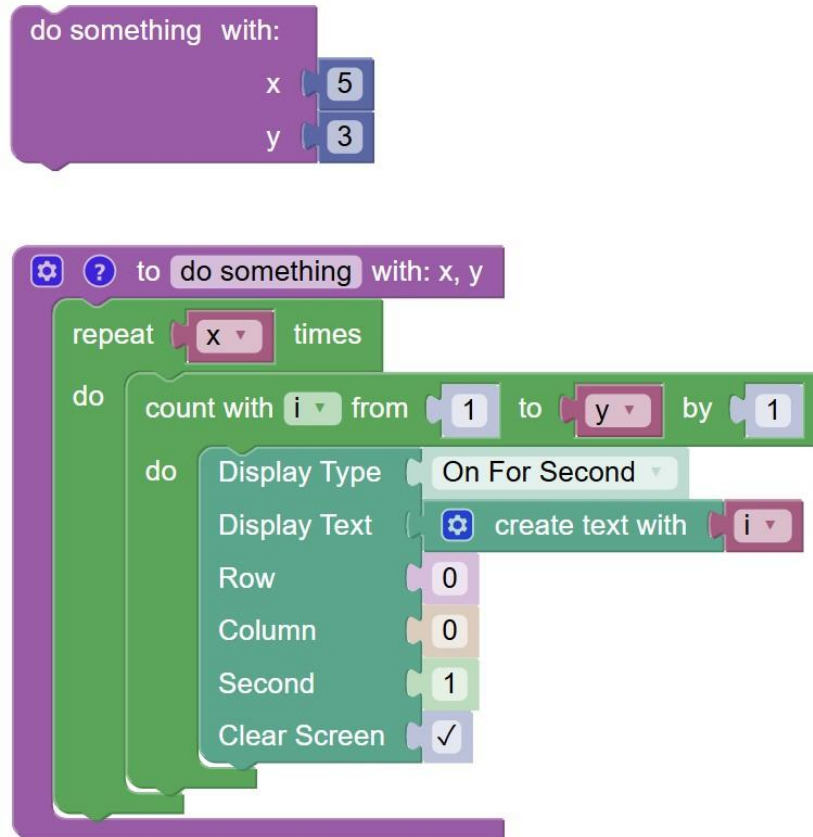
When the following code is run, what will be displayed?

- A) Hello, x
- B) The LabHello,
- C) Hello, TheLab
- D) Nothing, the program will crash



Exercise Q2

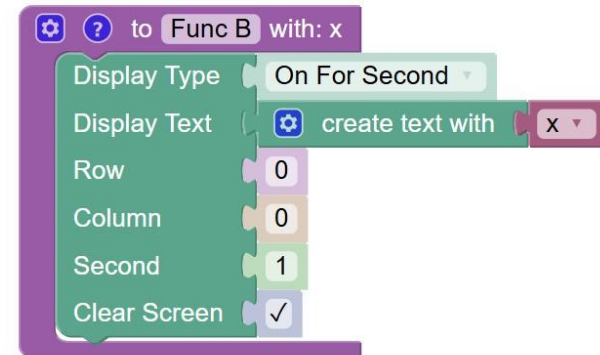
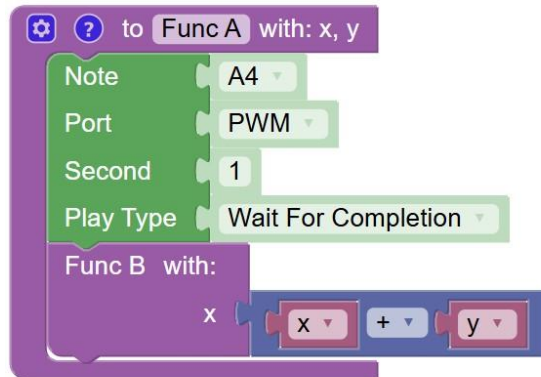
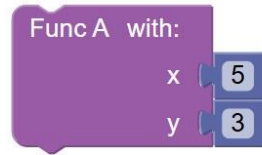
What happens when the following code is run?



- A) It will display 1 to 5 for three times.
- B) It will display 1 to 3 for five times.
- C) It will display i for fifteen times.
- D) The program will crash

Exercise Q3

What will be displayed when the following code is run?

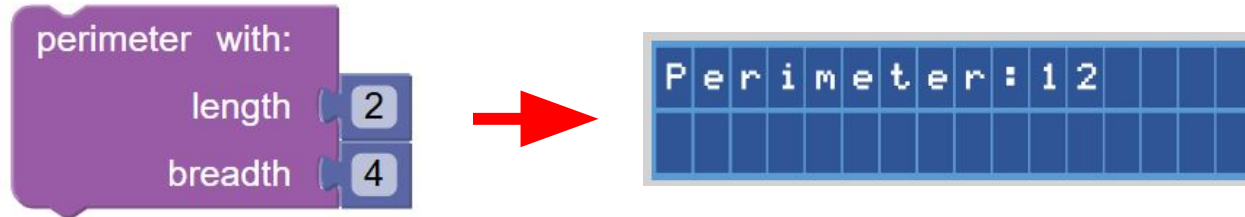


- A) 3
- B) 5
- C) 8
- D) The program will crash

Exercise Q4

Implement a function to take in the length and breadth of a rectangle and display its perimeter.
 $\text{perimeter} = 2 \times (\text{length} + \text{breadth})$.

Below shows a sample output.



Answers

Q1: C

"TheLab" is passed to the argument x, so the function will display "Hello, TheLab".

Q2: B

x is 5 and it will repeat 5 times the counting of 1 to y, where y is 3.

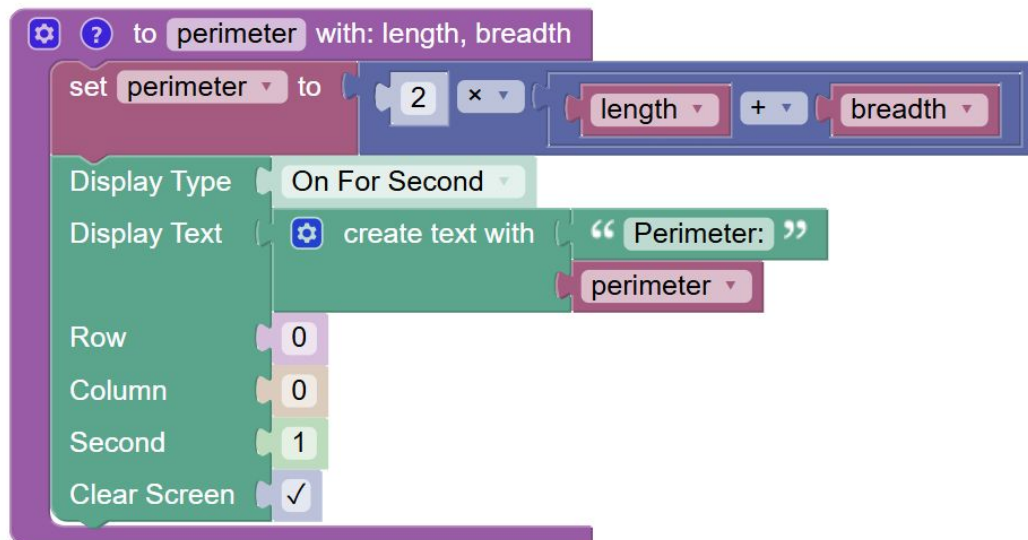
Q3: C

In Func A, x is 5 and y is 3.

$x + y = 8$ and is passed to x of Func B.

Hence, 8 is printed.

Q4:



3027 - Function Return

Learning

A Function can also **return** computed results instead of sending them to output modules like LCD or buzzer. The main program which calls the function will then decide what to do with the results.

In this unit, you'll explore how functions return results and how to capture them for use.

In class, we will explore the following:

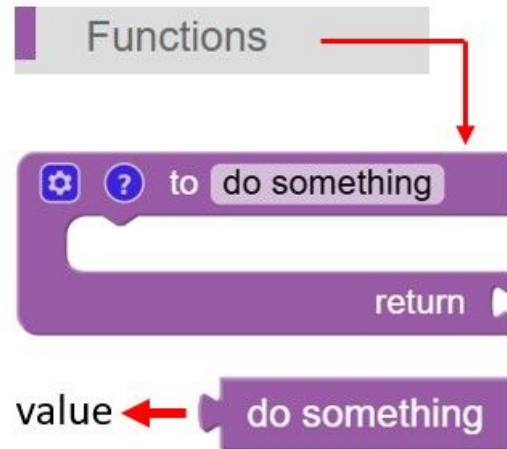
- Creating function that returns value.
- Applications including creating a Money Counter.



3027 - Function Return

Function returning a value

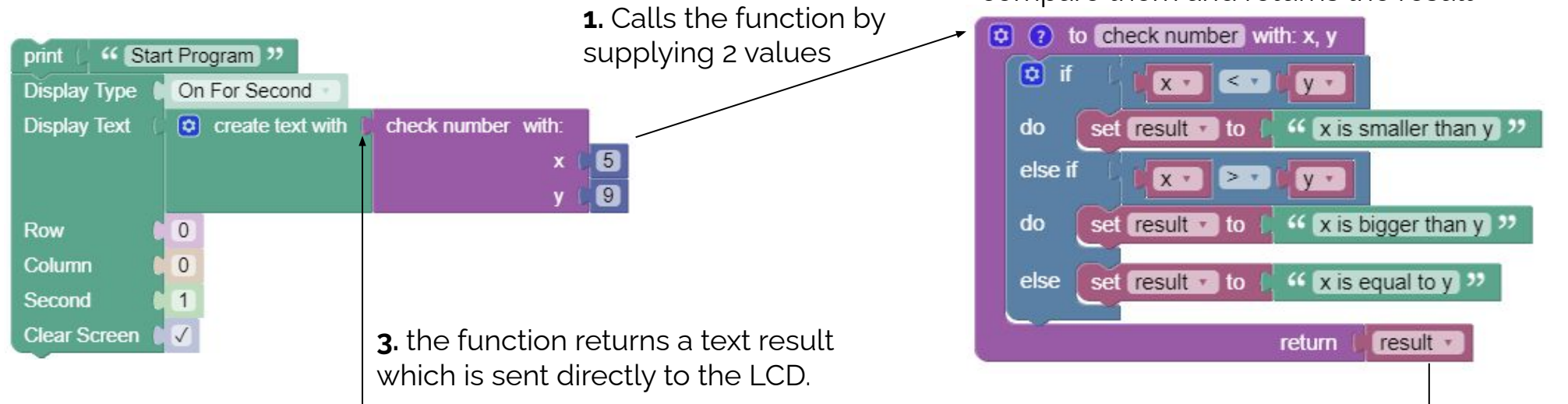
Use the function with return block to create functions that return values.



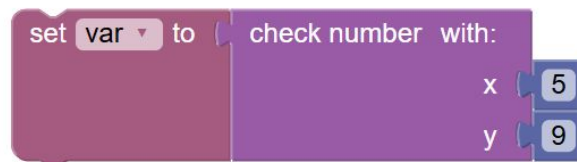
3027 - Function Return

Function returning a value

Below is an example of how you capture returned results.



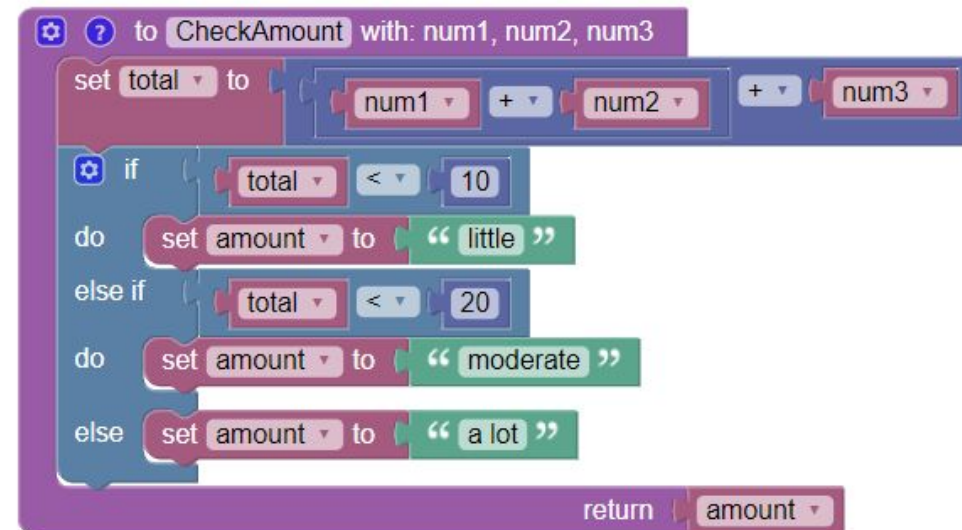
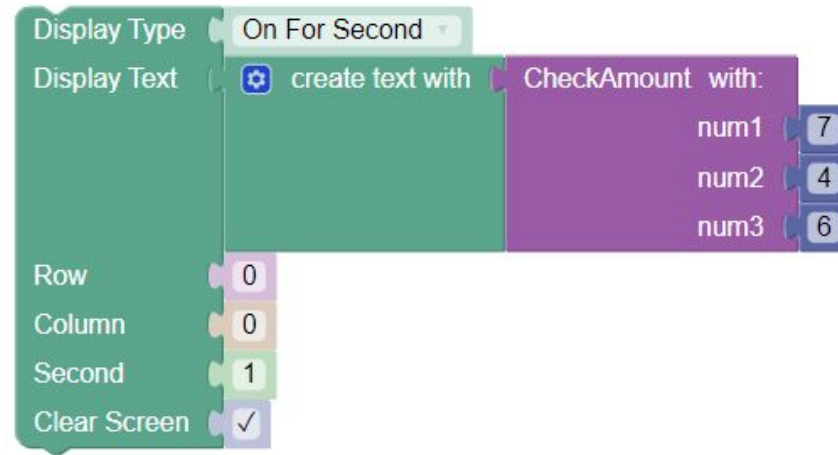
You can also save the results in a variable to use it.



Exercise Q1

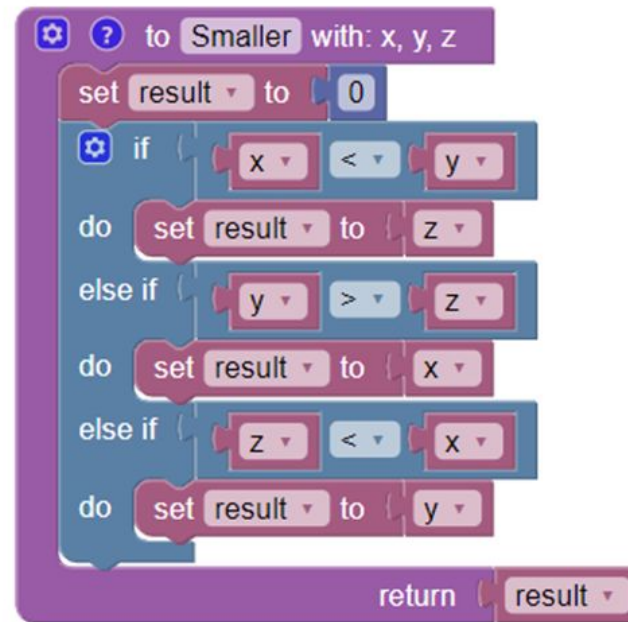
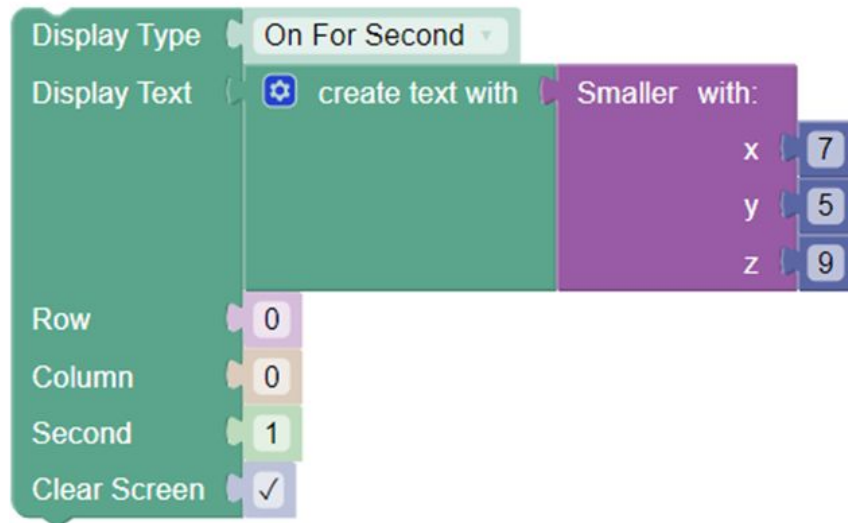
When the following code is run, what will be displayed?

- A) 7, 4, 6
- B) little
- C) moderate
- D) a lot



Exercise Q2

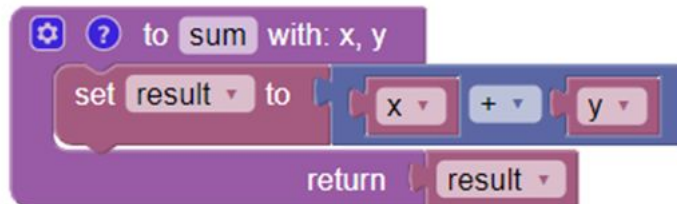
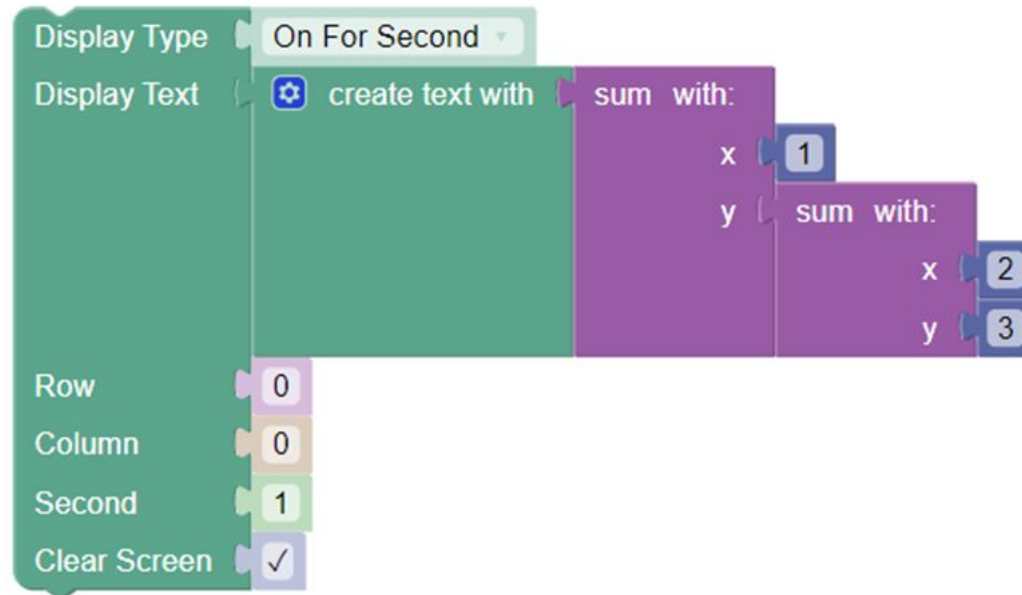
When the following codes are run, what will be displayed?



- A) 0
- B) 5
- C) 7
- D) 9

Exercise Q3

When the following codes are run, what will be displayed?

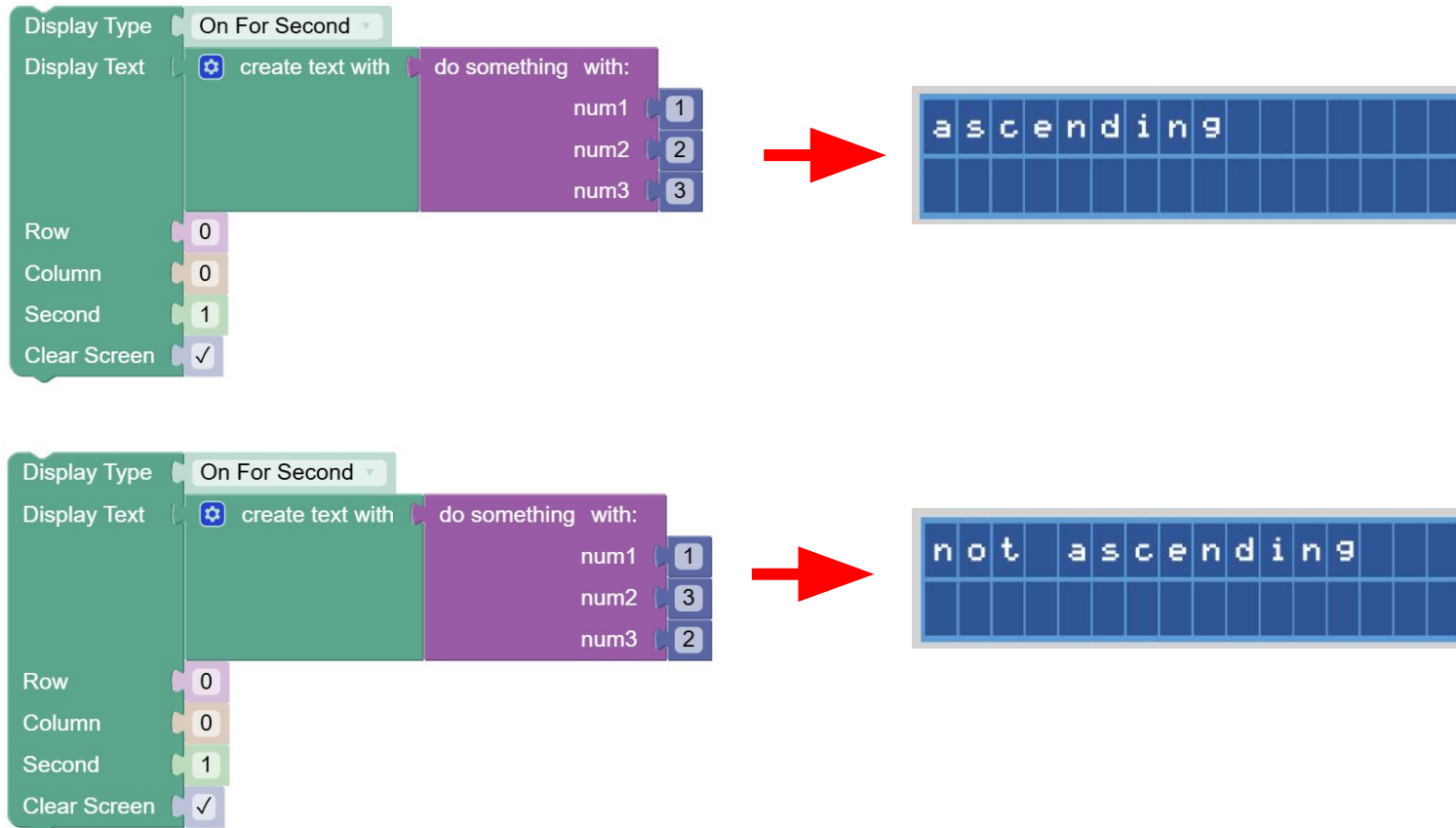


- A) 5
- B) 6
- C) 123
- D) 321

Exercise Q4

Create a function that compares 3 numbers and returns “ascending” or “not ascending”. The numbers are ascending if the first is smaller than the second and the second is smaller than the third.

Below shows some sample outputs.



Answers

Q1: C

When the function is called,

num1=7, num2=4 and num3=6.

total=num1+num2+num3=7+4+6=17

Total is not <10 but is <20 so "moderate" is displayed.

Q2: A

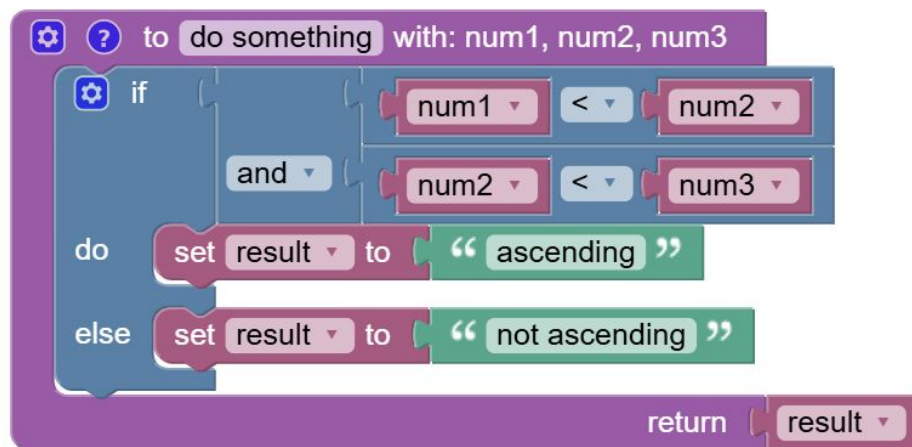
None of the if-conditions are true so 0 will be displayed.

Q3: B

The function is called twice sum(1, sum(2, 3)).

sum(2, 3) gives 5. So sum(1, 5) gives 6.

Q4:



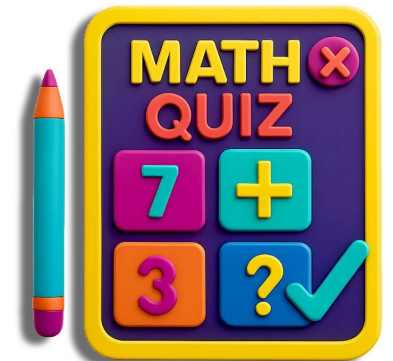
3028 - Function Applications

Learning

Having learnt about functions, you will apply your knowledge to develop a longer program with multiple functions.

In class, we will explore the following:

- Breaking down a larger program into parts and design it with multiple functions to perform various tasks through the development of a Math Quiz program.



Exercise Q1

With the advanced concepts of List and Function, your program can now be scalable and efficient. This is your last Blockly project practice, so give it your best shot!

To create a project, you need to:

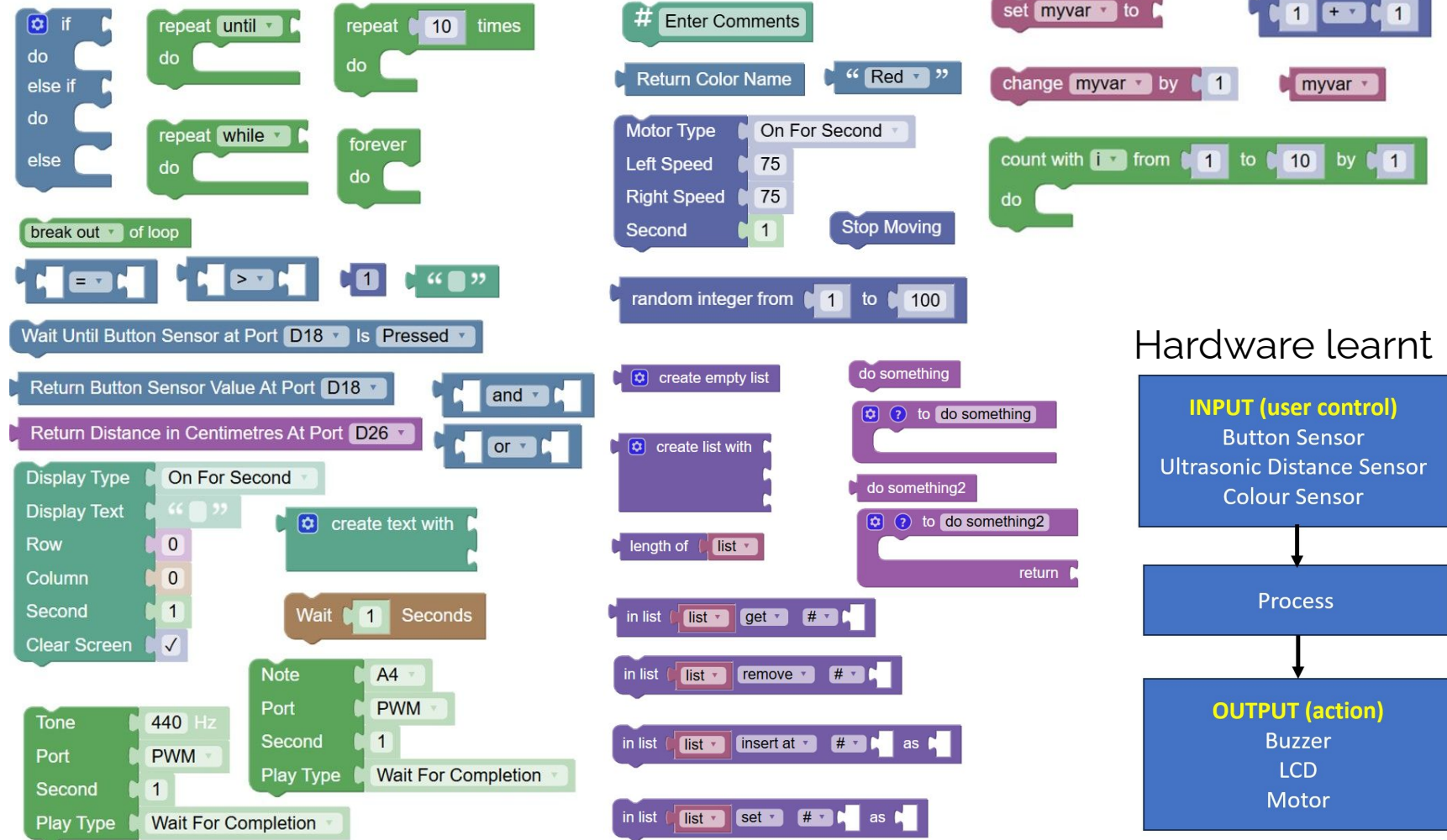
1. Understand individual **block usage**
you can revise the notes and do the exercises to build a strong foundation
2. Practice **combining and applying** them to solve problems
you need to be attentive and diligent in class to master it
3. Think of a concept and **outline the sequence** of tasks
you can get creative or use an idea you came across before
4. Implement the outline **one part at a time**
all big problems are solved in small steps

3029 - Intermediate 2 Project

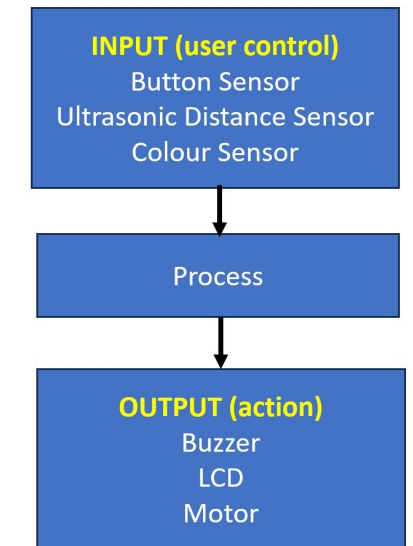
Learning

Mastering functions empowers you to design larger, more efficient programs with greater ease and flexibility. Now's your chance to showcase the full range of your coding skills. Let this final Blockly project be a testament to your growth and creativity!

Blocks learnt



Hardware learnt



END